

PUBLIC TECHNICAL BENCHMARK

DigitalOcean Inference Endpoint Technical Benchmark

Incomplete evidence report - 23-24 August 2026

DRAFT - INCOMPLETE COVERAGE - NOT FOR PUBLICATION

12 reported endpoints | 12 endpoint profiles | capacity, latency, limits, quality, cost, and coverage

Observed facts, calculated estimates, and interpretations are labelled separately. Missing evidence is shown explicitly.

1. Executive conclusion

This report maps the measured operating envelope of the DigitalOcean-hosted endpoints in the supplied public bundle. It separates what was observed from what was documented, distinguishes service reliability from task quality, and labels every missing or statistically weak region explicitly. The endpoint pages are the decision surface: each uses the same inventory, coverage, capacity, limits, quality, cost, and limitations structure.

What can this report support?

Plain language: It can guide workload-specific starting limits and identify where more measurement is required.

Technical definition: An AIMD capacity claim requires three explicit healthy confirmation epochs at one exact matched-cell rate, and is reported as a bracket or a right-censored lower bound. A sustained claim additionally requires the independent soak blocks. The report does not convert sparse observations into an SLA.

Exact endpoints represented	12
Request-level observations	41595
Request-attributed estimated cost	\$189.14
Conservative campaign exposure	\$237.36
Campaign cost cap	\$400.00
Billing/credit HTTP 402 latch observed	Yes
Completed coverage cells	105
Planned coverage cells	192
Analysis generated	2026-08-24T11:29:25.565106+00:00
Analysis schema	digitalocean_direct_public_analysis_v3

Decision rule

Treat an AIMD number only as a confirmed healthy offered arrival rate for the exact endpoint, workload shape, streaming mode, token lengths, epoch duration, and concurrency regime shown. It is not completed goodput. Use the separately reported achieved-RPM interval and require a passing drain-inclusive soak before calling a rate sustained. This report does not manufacture operational headroom; choose it from your risk target, retain adaptive backoff, and rerun the matched cell when conditions change.

Metric correction

The old six-digit token-rate outliers were a timing-definition error

Plain language: They divided completion tokens by the gap between streamed chunks; one chunk may contain many tokens.

Technical definition: The legacy maximum was 292,454 TPS. It is now audit-only. The corrected post-TTFT proxy retains its full qualified range (maximum 704.57 TPS), while aggregate successful billed-token goodput is the headline throughput. Buffered TTFT, multi-choice per-sequence rates, and timing-unstable sub-100 ms intervals are explicitly censored. The 100 ms rule is a post-hoc measurement correction and does not remove requests from reliability, cost, token, quality, or aggregate-goodput accounting. No qualified extreme is silently trimmed.

Largest qualified timing observations

These are the largest observations that survive the declared timing rules. They are shown row by row so a reader can audit the token count and denominator; none is silently clipped or winsorized.

Endpoint / workload	Output tokens	TTFT s	Total s	Post-TTFT denominator s	Cache state	Qualified proxy tokens/s	Classification
qwen3.8-max Direct capability parameter	65,536	0.5	93.51	93.01	Cache miss observed	704.57	Valid ordinary
arcee-trinity-large-thinking Tool call exact	68	1.35	1.47	0.12	Not reported unknown	546.74	Valid ordinary
minimax-m2.5 Direct short exact	64	8.17	8.29	0.12	Not reported unknown	536.59	Valid ordinary
minimax-m2.5 Direct short exact	64	6.5	6.62	0.12	Not reported unknown	526.32	Valid ordinary
mimo-v2.5-pro Direct capability parameter	65,536	0.38	125.53	125.15	Cache hit observed	523.67	Valid ordinary
minimax-m2.5 Direct short exact	64	11.07	11.19	0.13	Not reported unknown	504.24	Valid ordinary
nemotron-3-ultra-550b Parameter interactions	64	1	1.13	0.13	Not reported unknown	490.12	Valid ordinary
minimax-m2.5 Direct short exact	64	7.98	8.12	0.14	Not reported unknown	468.38	Valid ordinary
deepseek-v4-flash-0731 Direct mixed tool	69	1.12	1.28	0.16	Not reported unknown	429.16	Valid ordinary
minimax-m2.5 Short input / short output	64	2.11	2.27	0.16	Cache hit observed	398.89	Valid ordinary

Measured capacity navigation

This compact matrix is a navigation aid, not a provider ranking. A two-minute pass is shown before AIMD evidence; otherwise a rate is labelled as a healthy offered-rate bracket/lower bound or as exploratory. Exact token counts, concurrency, achieved-rate intervals, latency, TPM, failures, and recovery evidence remain in the matched endpoint profile. When qualified runs disagree, the observed range across runs is shown rather than silently selecting the fastest run.

Endpoint	Short / short	32K / short	Short / long	Mixed
arcee-trinity-large-thinking	2-min offered-rate fail 60 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 60 RPM	2-min offered-rate fail 24 RPM
deepseek-v4-flash-0731	2-min healthy offered 60 RPM; fail 960	2-min offered-rate fail 24 RPM	2-min offered-rate fail 96 RPM	2-min offered-rate fail 96 RPM
gemma-4-31B-it	2-min healthy offered 60 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM
glm-5.2	2-min offered-rate fail 60 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 6 RPM
kimi-k3	2-min offered-rate fail 60 RPM	2-min offered-rate fail 36 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 192 RPM
minimax-m2.5	2-min offered-rate fail 240 RPM	2-min offered-rate fail 21 RPM	2-min offered-rate fail 21 RPM	2-min offered-rate fail 12 RPM
mimo-v2.5-pro	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 12 RPM
nemotron-3-ultra-550b	2-min offered-rate fail 60 RPM	2-min offered-rate fail 6 RPM	2-min offered-rate fail 60 RPM	2-min offered-rate fail 24 RPM
nvidia-nemotron-3-super-120b	Healthy offered 60-120 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 6 RPM	Exploratory healthy 48 RPM
openai-gpt-oss-120b	2-min offered-rate fail 240 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 24 RPM
qwen3.5-397b-a17b	2-min offered-rate fail 960 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 12 RPM	Exploratory healthy 48 RPM
qwen3.8-max	2-min healthy offered 60 RPM	2-min offered-rate fail 12 RPM	2-min offered-rate fail 24 RPM	2-min offered-rate fail 6 RPM

Engineering disposition by measured workload

A two-minute pass means only that the exact soak passed at the shown offered load; it is not a production recommendation or SLA. Restricted means the soak produced a measured failure. Transport impaired means the low-load gate failed. Exploratory means only short AIMD confirmation exists. Apply only to the exact contract in the endpoint profile and retain retry/backoff.

Endpoint	Short / short	32K / short	Short / long	Mixed
arcee-trinity-large-thinking	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 24 RPM
deepseek-v4-flash-0731	2-min soak passed 60 offered RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 96 RPM	Restricted 2-min fail at 96 RPM
gemma-4-31B-it	2-min soak passed 60 offered RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM
glm-5.2	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 6 RPM
kimi-k3	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 36 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 192 RPM
minimax-m2.5	Restricted 2-min fail at 240 RPM	Restricted 2-min fail at 21 RPM	Restricted 2-min fail at 21 RPM	Restricted 2-min fail at 12 RPM
mimo-v2.5-pro	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 12 RPM
nemotron-3-ultra-550b	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 6 RPM	Restricted 2-min fail at 60 RPM	Restricted 2-min fail at 24 RPM
nvidia-nemotron-3-super-120b	Transport impaired baseline gate	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 6 RPM	Transport impaired baseline gate
openai-gpt-oss-120b	Restricted 2-min fail at 240 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 24 RPM
qwen3.5-397b-a17b	Restricted 2-min fail at 960 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 12 RPM	Transport impaired baseline gate
qwen3.8-max	2-min soak passed 60 offered RPM	Restricted 2-min fail at 12 RPM	Restricted 2-min fail at 24 RPM	Restricted 2-min fail at 6 RPM

2. Definitions and measurement method

Each term is explained twice: first as an engineering intuition, then as the precise calculation used to support comparisons.

Time to first token (TTFT)

Plain language: How long a streaming caller waits before any answer text arrives.

Technical definition: TTFT is the monotonic-clock interval from request dispatch to receipt of the first non-empty response token. It includes network, queue, and end-to-end prefill time; it is not a direct server-side prefill measurement.

End-to-end latency

Plain language: How long the whole request takes from send to completion or terminal failure.

Technical definition: Latency is terminal timestamp minus dispatch timestamp. Percentiles use independent requests or epochs as the sampling unit and include failures in reliability summaries.

Aggregate output goodput

Plain language: How many successfully billed answer tokens finish per wall-clock second or minute.

Technical definition: Headline output goodput is successful server-reported completion tokens divided by the complete epoch, soak block, or source-active wall-clock interval. It includes reasoning tokens when the provider bills and reports them.

Post-TTFT service-output proxy

Plain language: A conservative per-request view of output delivery after streaming begins.

Technical definition: Billed completion tokens / (request end - streamed TTFT). This includes network and response buffering and is not direct decoder speed. Sub-100 ms intervals are timing-unstable and are explicitly censored from this rate but retained in aggregate goodput. The first-to-last SSE-event span is audit-only because one event can contain many tokens.

Cache stratum

Plain language: Whether DigitalOcean explicitly reported reused prompt tokens, a miss, or no cache counter.

Technical definition: TTFT is never pooled across observed cache hits, observed misses, and unknown cache state. Prompt tokens / TTFT is headline-eligible only for an explicit cache miss. Missing counters are not treated as zero.

Effective input and output TPM

Plain language: How many useful prompt and answer tokens the endpoint successfully handles per minute.

Technical definition: Effective input TPM = successful server-reported prompt tokens / elapsed minutes. Effective output TPM = successful server-reported completion tokens / elapsed minutes. Offered and failed tokens are reported separately.

Goodput

Plain language: Throughput that actually produced usable answers.

Technical definition: Goodput counts successful, non-truncated requests that pass the workload validity rule. Quality-adjusted goodput additionally requires deterministic task correctness or a preregistered quality threshold.

AIMD capacity sweep

Plain language: The test raises load while healthy and backs off sharply after congestion.

Technical definition: Additive increase, multiplicative decrease uses open-loop arrivals with a separate concurrency ceiling. Three separated healthy confirmation epochs establish a confirmed healthy offered arrival rate, not completed or sustained capacity. Achieved drain-inclusive goodput remains a separate interval; an observed maximum without confirmations is exploratory.

95% confidence interval

Plain language: A range showing how uncertain the estimate is across independent repetitions.

Technical definition: Intervals use independent requests or load epochs, never individual output tokens. Sparse tail percentiles are marked exploratory; p99 is omitted unless roughly 1,000 relevant observations support it.

Buffered response

Plain language: A non-streaming request returns the whole answer at once.

Technical definition: Buffered calls expose full-response latency but not token-level TTFT. Their response-arrival timestamp is censored from TTFT, prefill, and post-TTFT per-sequence curves.

Statistical methodology

Latency and request metrics treat independent requests as observations; saturation capacity treats independent load epochs as observations. Endpoint order and comparable task instances should be randomized or interleaved, while each saturation sweep remains isolated. Confidence intervals describe sampling uncertainty in the measured regime, not future provider guarantees. Service errors, timeouts, refusals, malformed responses, truncations, and retries remain in the denominator appropriate to the metric.

- **Confidence level:** 95.0%
- **Bootstrap resamples:** 10000
- **Reproducibility seed:** 20260823
- **Low-load sampling unit:** request_id
- **AIMD sampling unit:** epoch_id

- **Soak sampling unit:** analysis_block_id
- **Paired-quality sampling unit:** quality_pair_id
- **Recovery sampling unit:** phase_id
- **Success-rate interval:** Wilson score interval
- **Low-load continuous-metric interval:** seeded request percentile bootstrap
- **AIMD interval:** seeded percentile bootstrap after one estimate per epoch
- **Minimum observations before reporting p99:** 1000
- **Output tokens treated as independent samples:** No
- **Soak interpretation:** Two-minute soak evidence is reported separately and never relabelled as an AIMD confirmation epoch
- **Soak-block dependence caveat:** Four contiguous 30-second analysis blocks preserve their exact analysis_block_id identities; Student-t block intervals are exploratory and do not model serial correlation

End-to-end prefill proxy

Plain language: TTFT shows the caller's total wait before generation begins.

Technical definition: Unless the service exposes authenticated server timing, prompt-processing speed is inferred only as an end-to-end proxy from TTFT versus realized server input tokens. It must not be described as direct server-side prefill throughput.

3. Cross-endpoint inventory and coverage

Endpoint	Version / region	Context / output	Capabilities
arcee-trinity-large-thinking	Not reported	128000 / 32000	cache, reasoning, text only
deepseek-v4-flash-0731	Not reported	1048576 / 1048576	cache, reasoning, text only
gemma-4-31B-it	Not reported	256000 / 8192	cache, text only
glm-5.2	Not reported	262144 / 262144	tools, JSON, cache, reasoning, text only
kimi-k3	Not reported	Not reported	vision, cache
minimax-m2.5	Not reported	65536 / 65536	cache, text only
mimo-v2.5-pro	Not reported	262144 / 262144	tools, JSON, cache, reasoning, text only
nemotron-3-ultra-550b	Not reported	131072 / 131072	text only
nvidia-nemotron-3-super-120b	Not reported	1000000 / 32768	cache, text only
openai-gpt-oss-120b	Not reported	128000 / 4096	cache, reasoning, text only
qwen3.5-397b-a17b	Not reported	131072 / 131072	cache, text only
qwen3.8-max	Not reported	1000000 / 262144	tools, JSON, vision, cache

Coverage accounting

Frozen endpoint x dimension cells	192
Completed or evidence-backed unsupported	105
Completed	97
Unsupported	8
Inconclusive	87
Untested / skipped	0

Time and cost ledger

Request-attributed estimated cost	\$189.14			
Conservative campaign exposure	\$237.36			
Campaign cost cap	\$400.00			
Billing/credit HTTP 402 latch observed	Yes			
Carried prior at first reconciled stage	52.892208102609274			
Requests with attributed cost	41,595 / 41,595			
Stage	UTC window	Prior	Final	Increment
AIMD + carried breadth reconciliation	2026-08-23 14:33 UTC to 2026-08-23 17:31 UTC	\$52.89	\$72.07	\$19.18
Independent two-minute soak	2026-08-23 18:31 UTC to 2026-08-23 21:24 UTC	\$72.07	\$111.04	\$38.97
Capability envelope	2026-08-23 21:41 UTC to 2026-08-23 21:47 UTC	\$111.04	\$125.06	\$14.02
Context envelope	2026-08-23 21:53 UTC to 2026-08-23 22:08 UTC	\$125.06	\$179.09	\$54.02
AIMD + carried breadth reconciliation	2026-08-24 04:36 UTC to 2026-08-24 06:48 UTC	\$179.09	\$197.84	\$18.75
Direct completion	2026-08-24 06:58 UTC to 2026-08-24 10:31 UTC	\$197.84	\$237.36	\$39.52

Request-attributed estimated cost sums normalized request usage. Conservative campaign exposure is the authoritative cumulative guard and retains reservations for incomplete, error, and unknown outcomes; the two values overlap, are not additive, and are not interchangeable. Each stage is checked against the cap in force for that stage; any owner-authorized cap increase is preserved explicitly in the ledger.

A billing/credit HTTP 402 latch was observed in the terminal context-envelope stage. Later context cells were therefore censored or left inconclusive; the latch is not evidence that a model capability is unsupported.

Explicit measurement exclusions

These subtests were excluded by the capability campaign manifest. They remain untested in the coverage ledger and cannot be interpreted as supported, unsupported, completed, or implicitly covered by a nearby test.

Explicitly untested measurement	Coverage dimension	Applies to	Reason
Adaptive tool +1 boundary discovery	tool_calling	12 endpoints	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	12 endpoints	a separate recovery experiment, not part of this no-retry capability measurement lane

4. Capacity, latency, quality, and cost comparisons

Every figure is a derived public view of the same endpoint and workload identities used in the tables. Categorical workload shapes are not connected as if they formed a continuous curve. Token axes should use realized server-reported token counts; requested anchors are labels, not substitutes for realized usage.

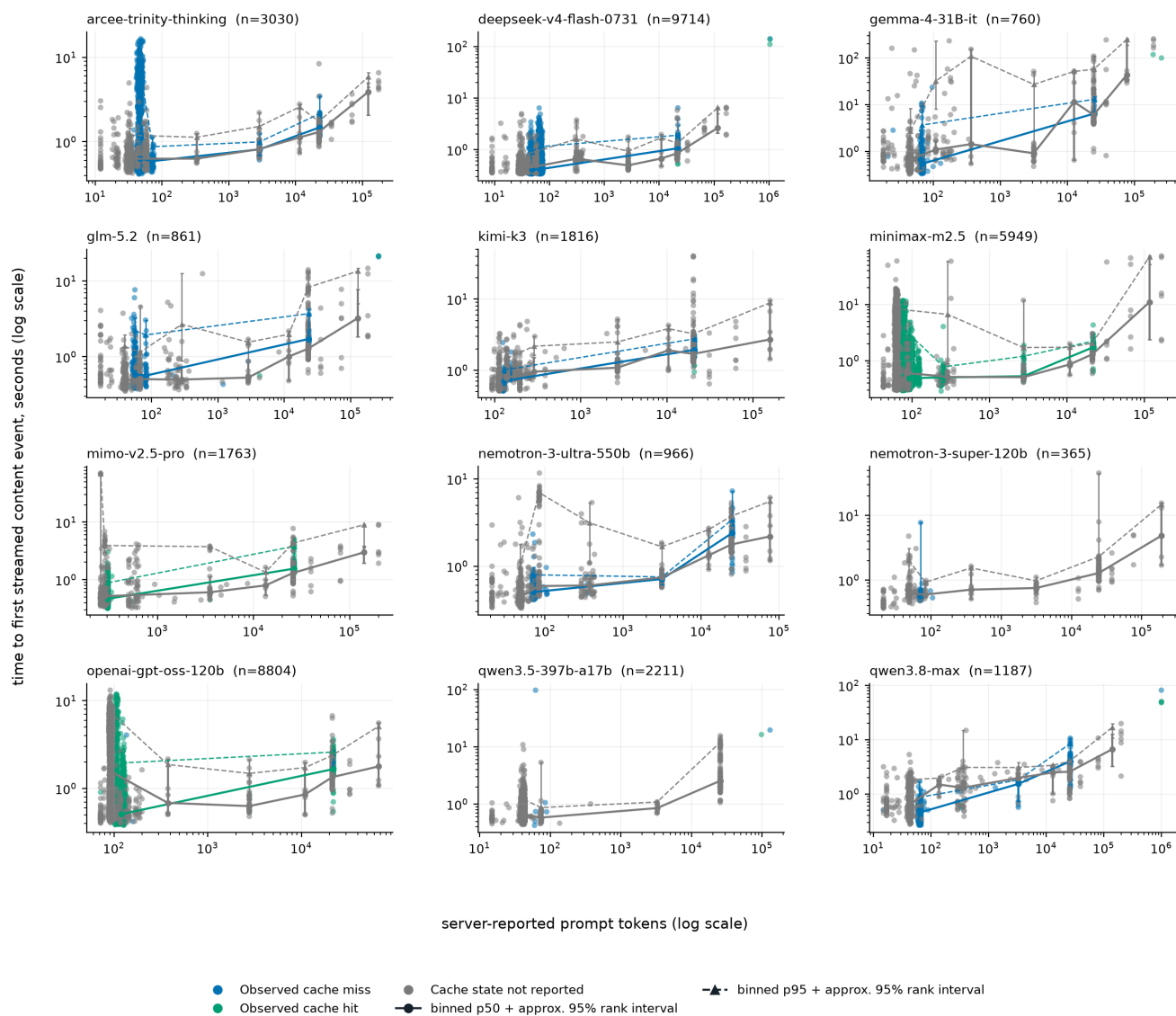
Coverage Status Matrix



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

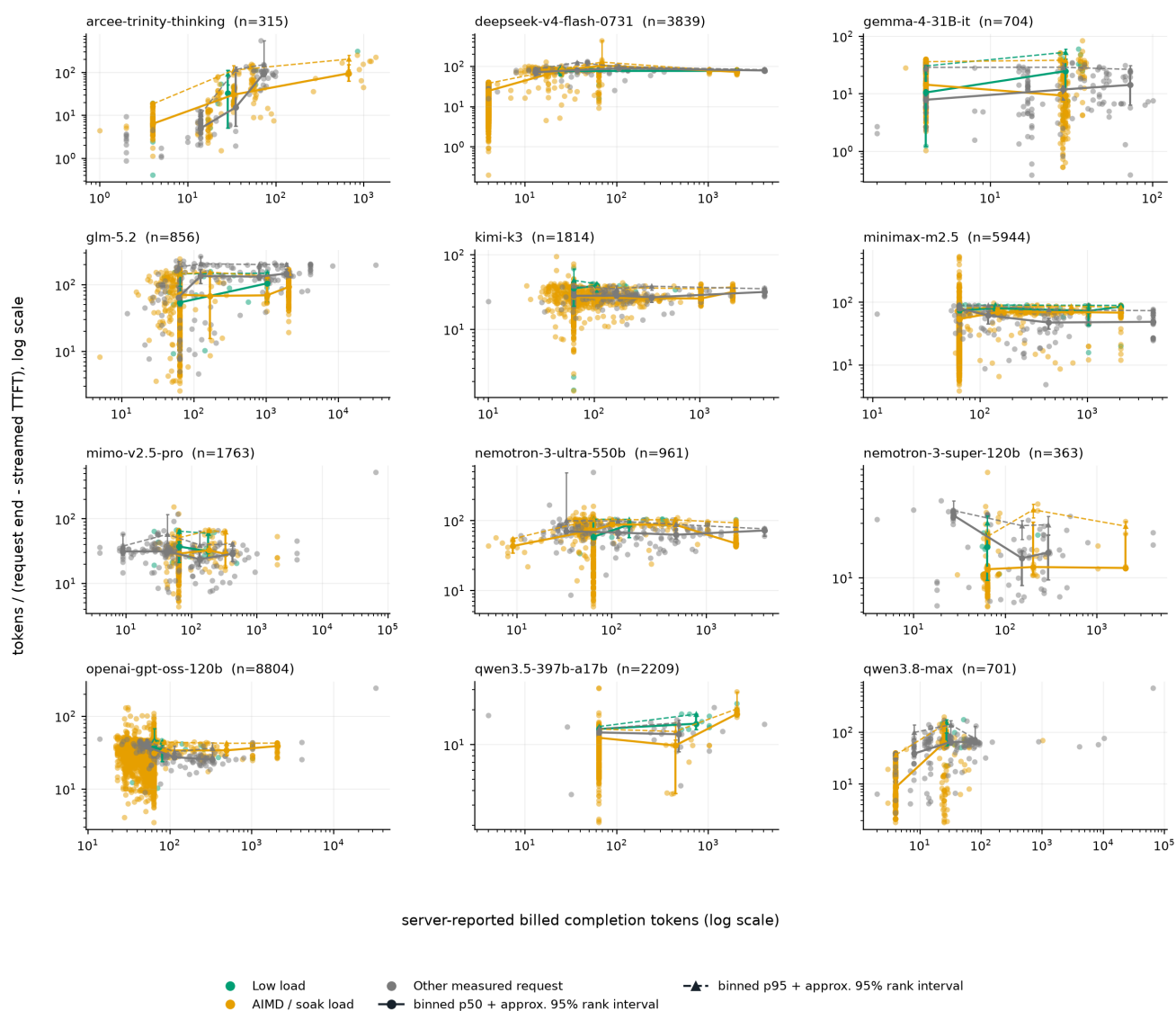
TTFT Input Cache Strata

Streaming TTFT versus server-reported input length, stratified by cache state



Output Post TTFT Proxy

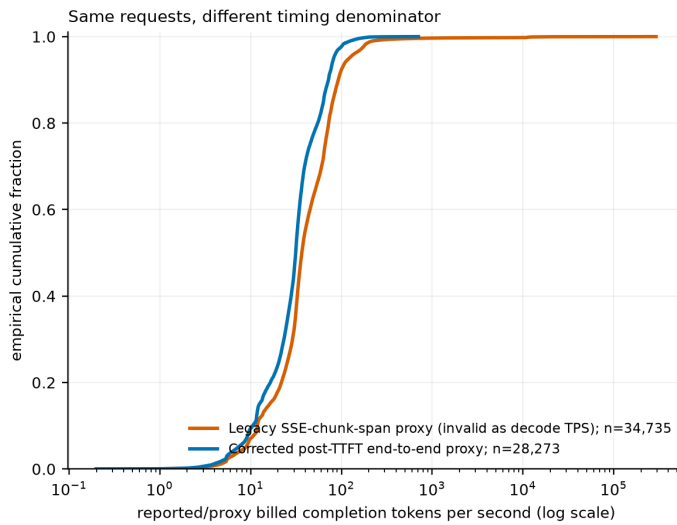
Billed completion-token service-output proxy versus realized output length



The y-axis is billed completion tokens divided by request end minus streamed TTFT. It is an end-to-end delivery proxy, not server decode speed. Multi-choice responses are excluded from this per-sequence view.

Metric Outlier Audit

Metric audit: legacy chunk timing versus corrected client-observed proxy



Why the old extremes were not decoder measurements

Legacy observations: 34,735
 Legacy $\geq 1,000$ tokens/s proxy: 122
 Legacy $\geq 10,000$ tokens/s proxy: 83
 Legacy $\geq 100,000$ tokens/s proxy: 9
 Legacy maximum: 292,454 tokens/s proxy

Corrected observations: 28,273
 Sub-100 ms intervals explicitly censored: 6,451
 Corrected median: 31.35 tokens/s proxy
 Corrected maximum retained: 704.57 tokens/s proxy

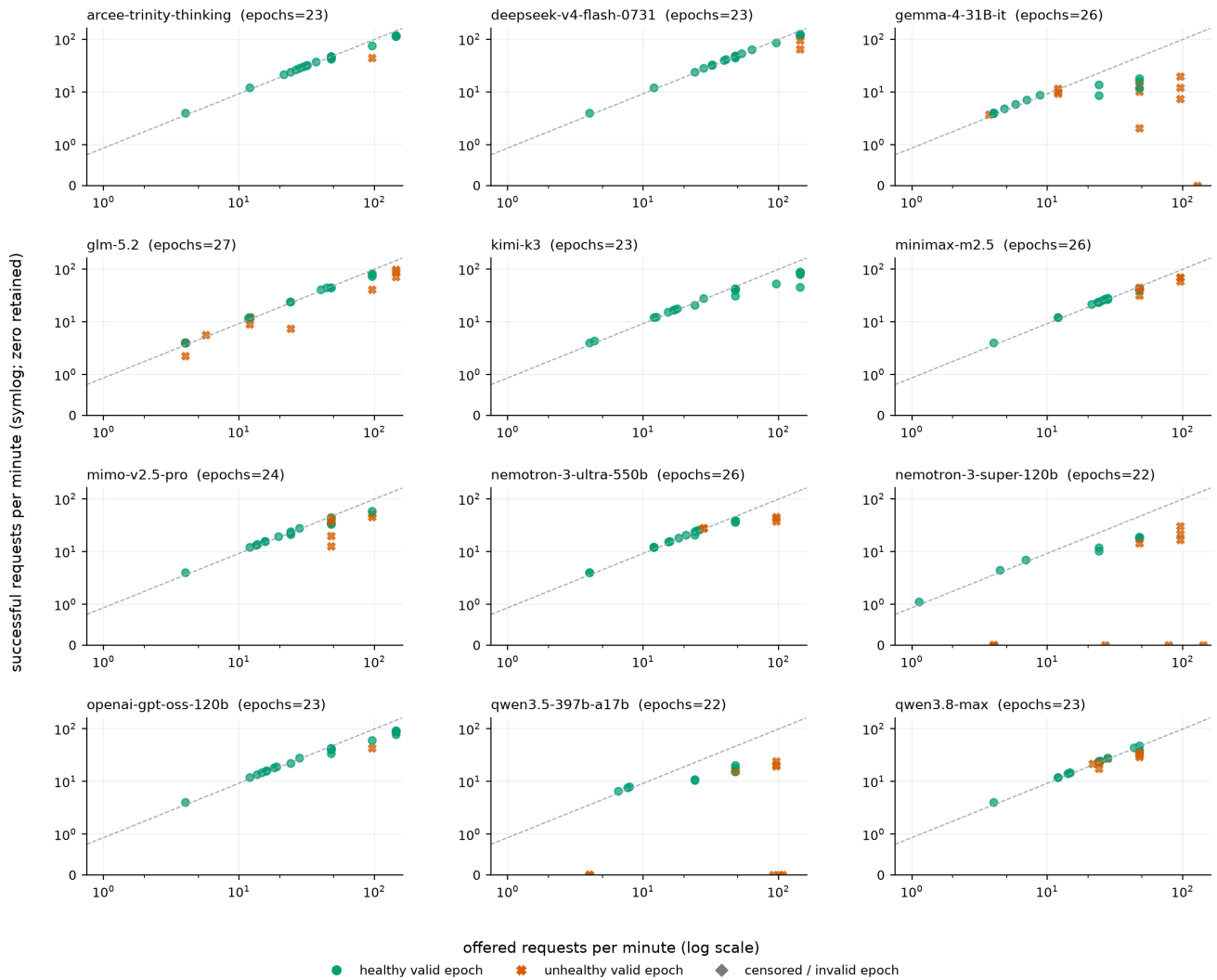
One-second denominator sensitivity audit:
 n=20,945; median=30.24
 p99=90.20; maximum=704.57 tokens/s proxy

SSE events are transport chunks, not tokens.
 No values were silently trimmed or winsorized.
 Sensitivity values are an audit, not a pooled endpoint comparison.

The red distribution is the rejected first-to-last SSE-event-span calculation. The blue distribution uses request end minus streamed TTFT. All valid extremes remain; invalid observations are censored with a reason rather than trimmed.

Capacity Input32k Short Matched Points

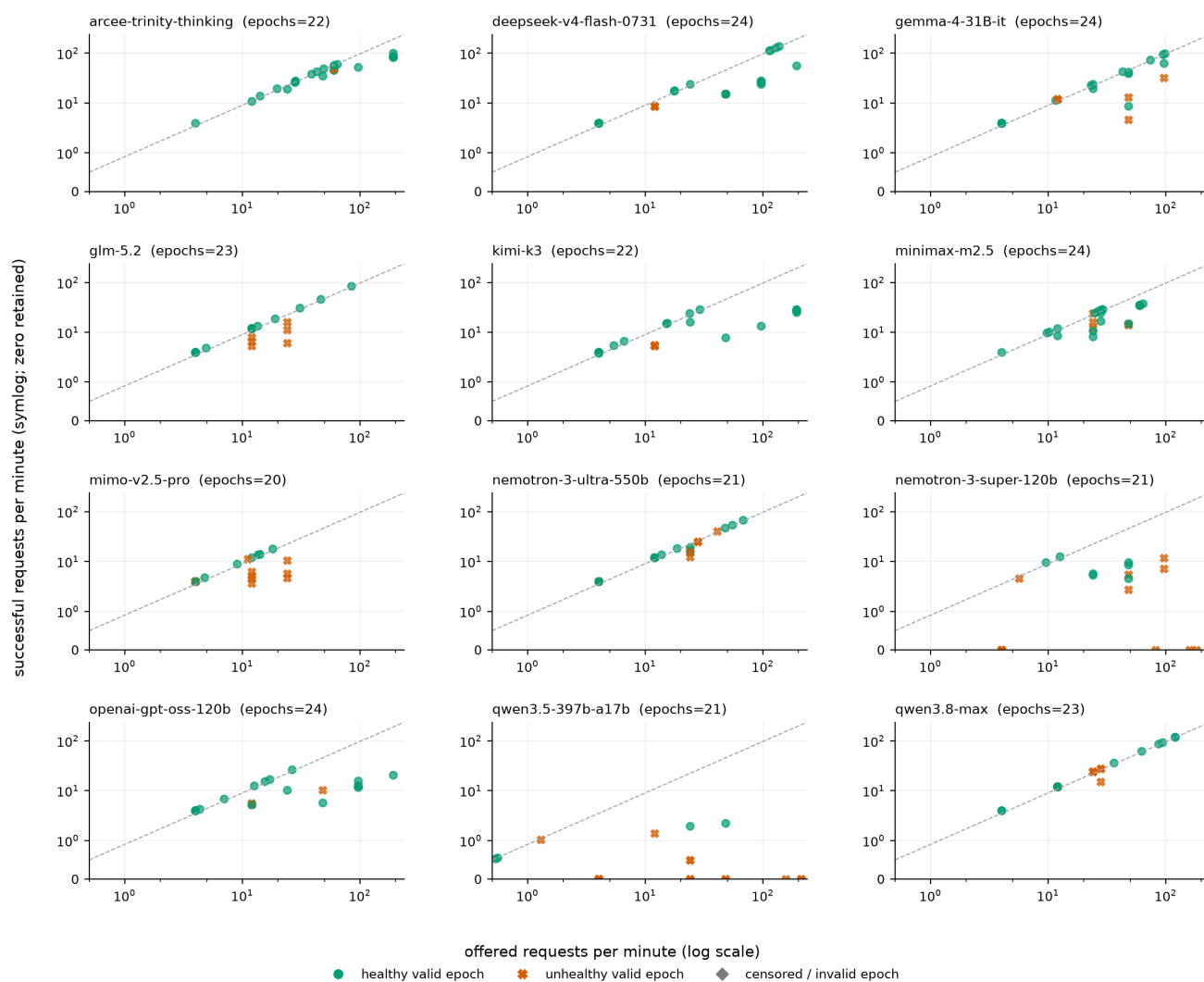
Open-loop capacity points: input32k short (matched workload only)



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

Capacity Mixed Matched Points

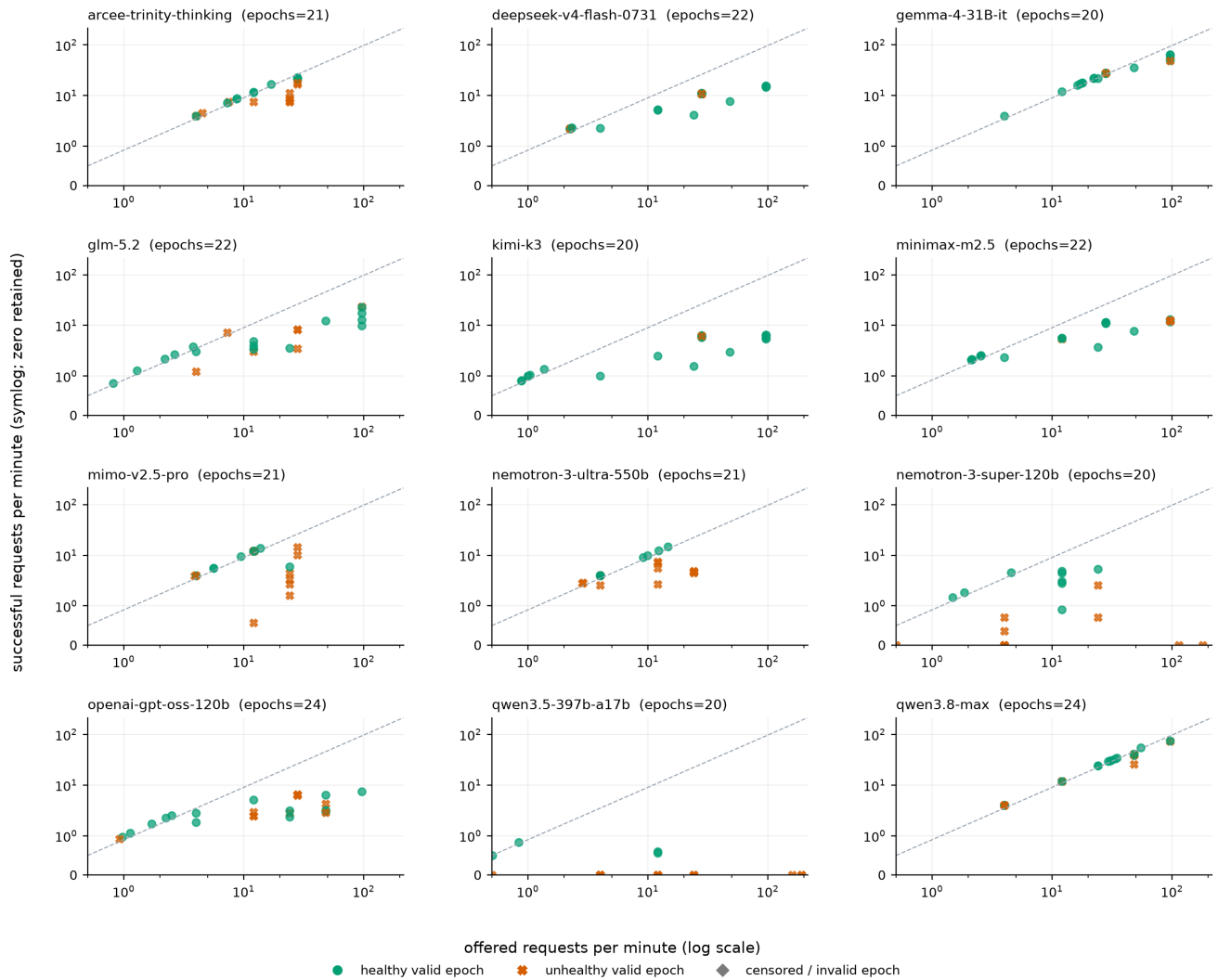
Open-loop capacity points: mixed (matched workload only)



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

Capacity Short Long Matched Points

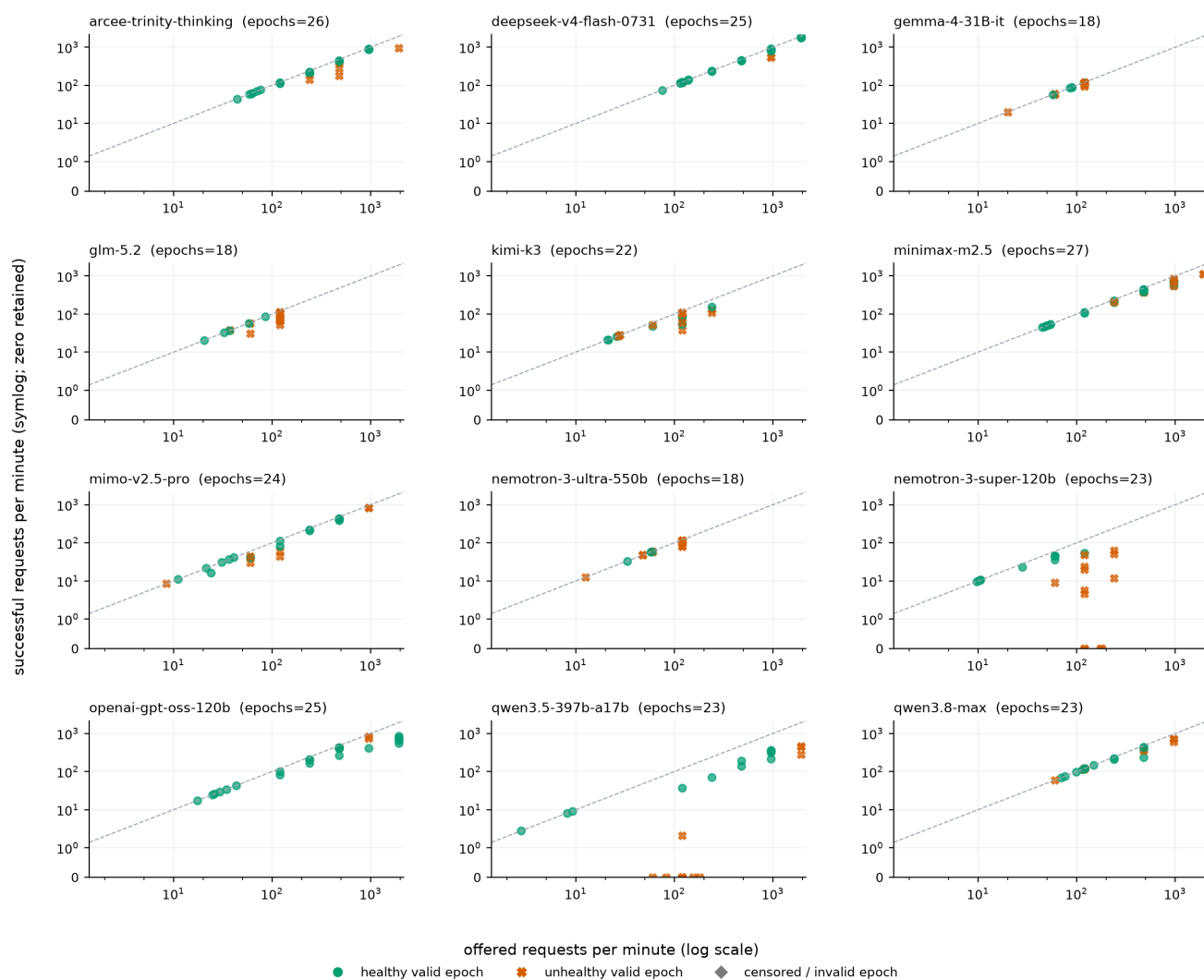
Open-loop capacity points: short long (matched workload only)



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

Capacity Short Short Matched Points

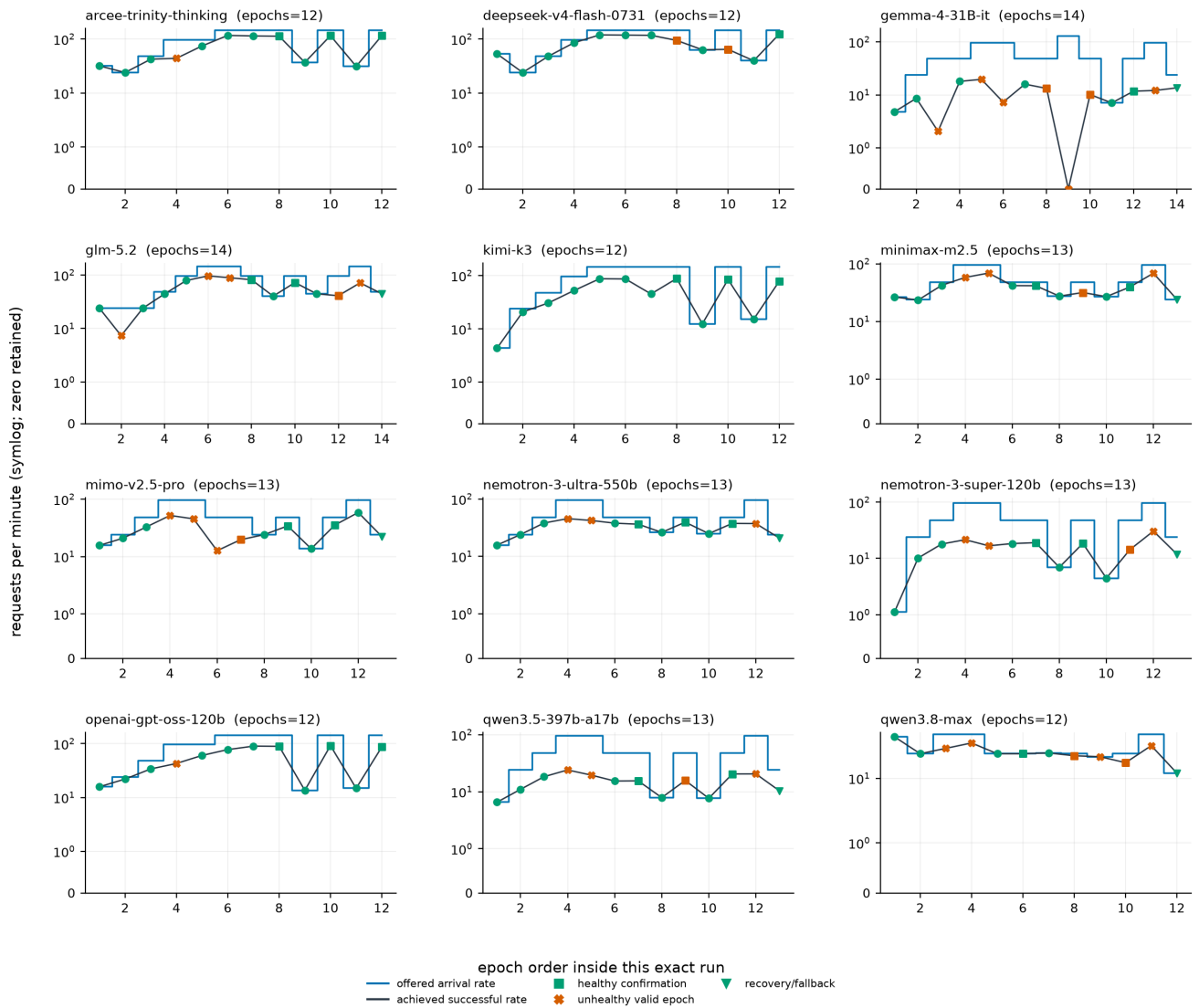
Open-loop capacity points: short short (matched workload only)



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

AIMD Controller Input32k Short Do Direct AIMD 20260823 Final

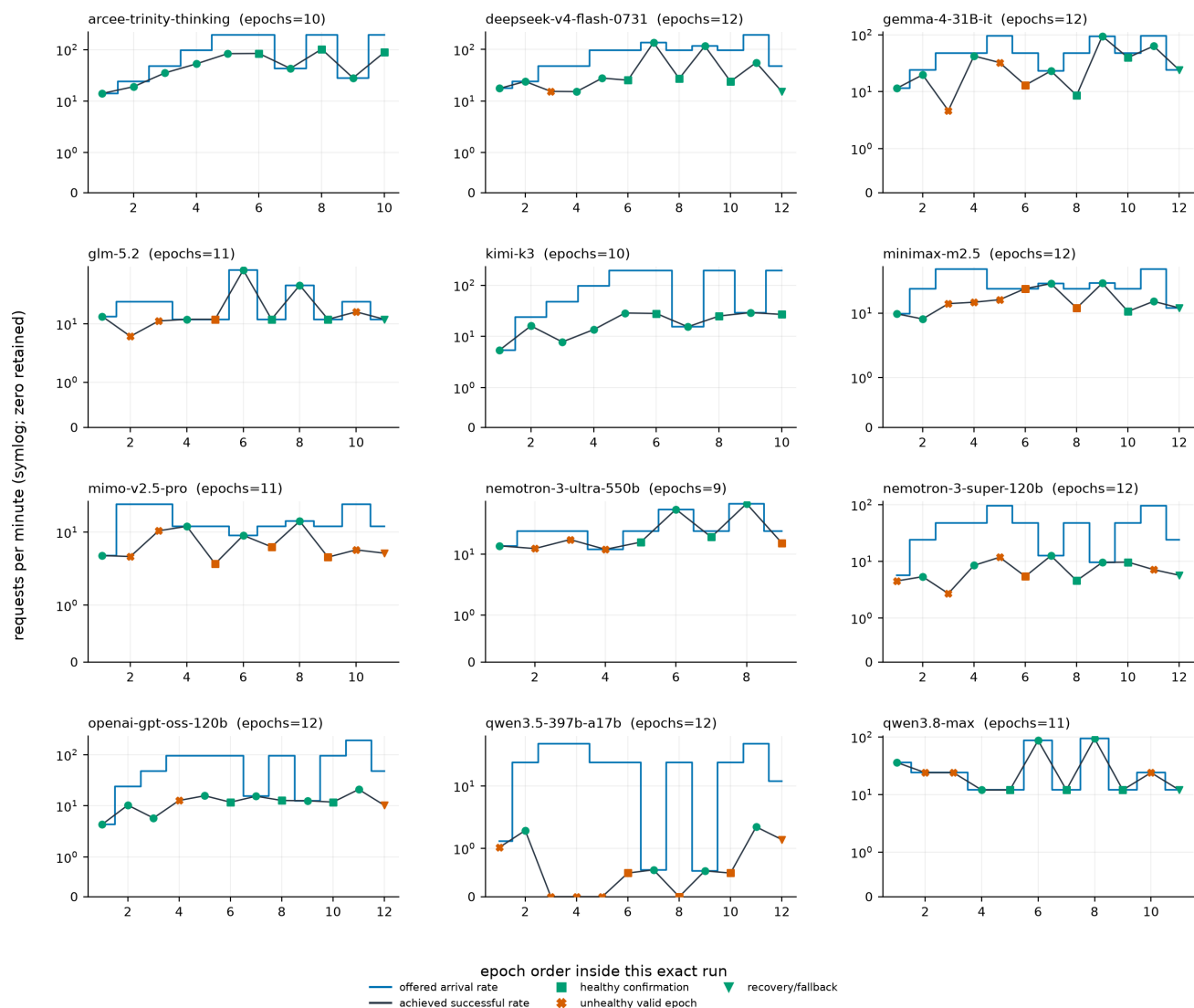
Chronological AIMD controller: input32k short
do-direct-aimd-20260823-final



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Mixed Do Direct AIMD 20260823 Final

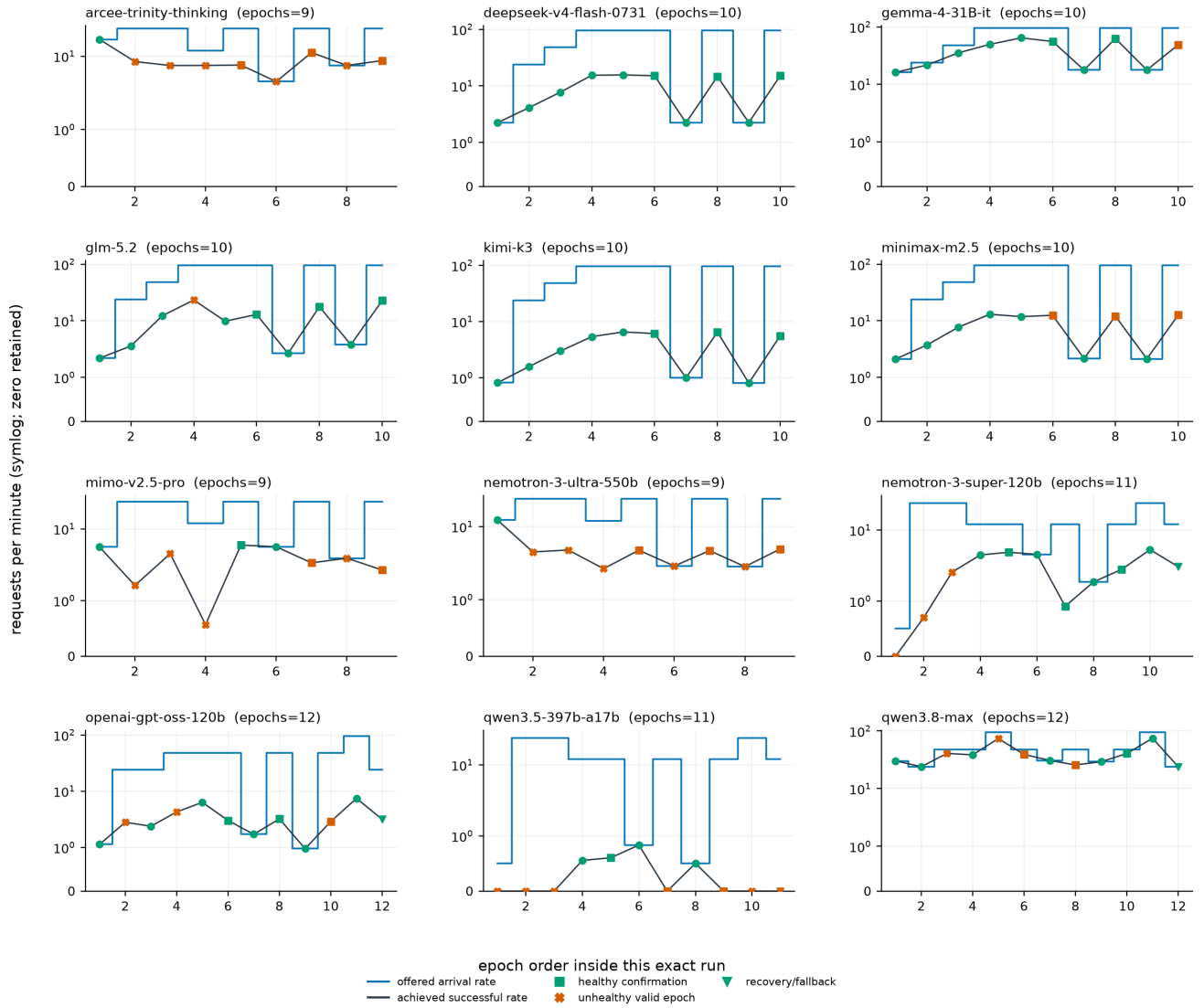
Chronological AIMD controller: mixed
do-direct-aimd-20260823-final



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Short Long Do Direct AIMD 20260823 Final

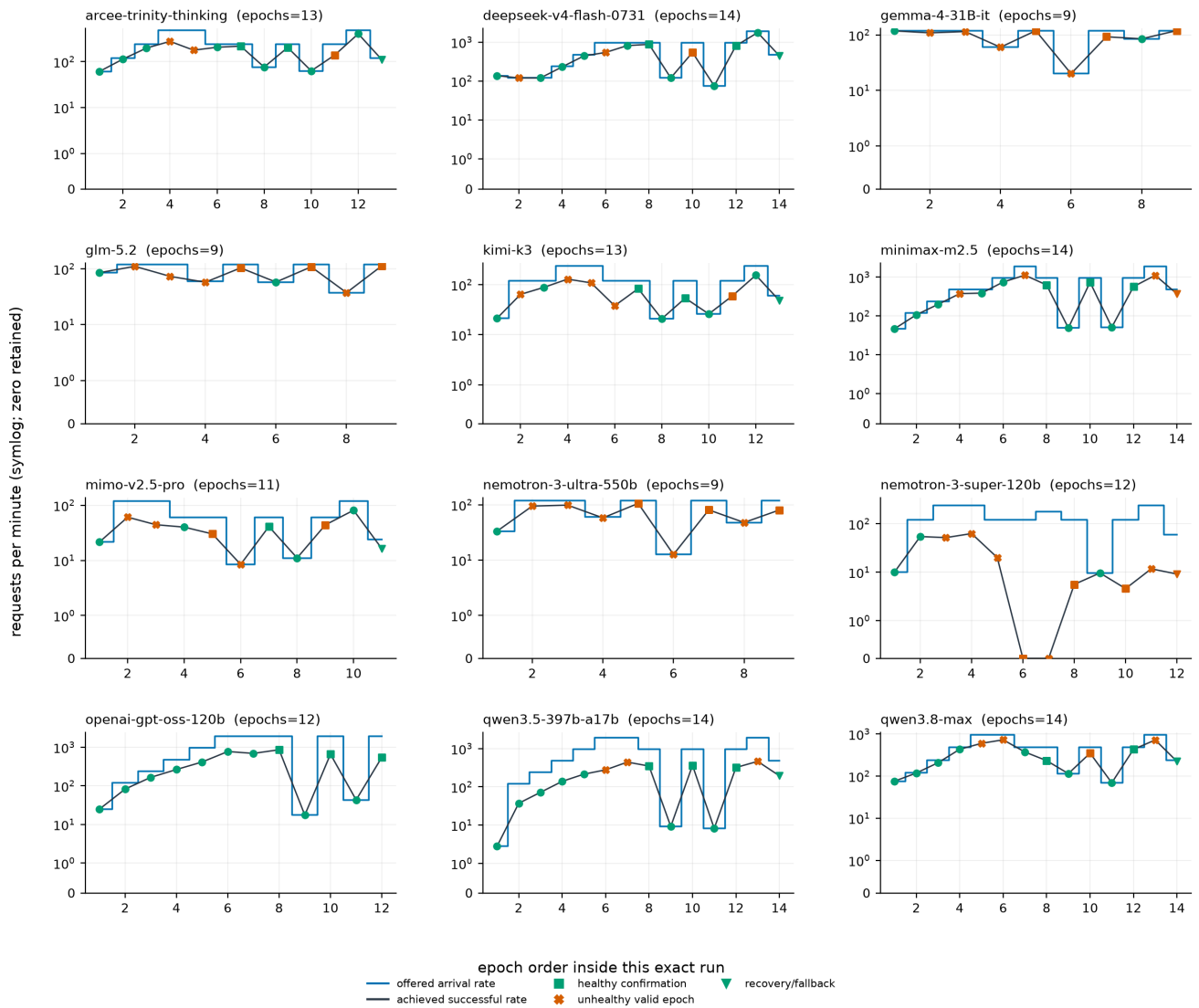
Chronological AIMD controller: short long
do-direct-aimd-20260823-final



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Short Short Do Direct AIMD 20260823 Final

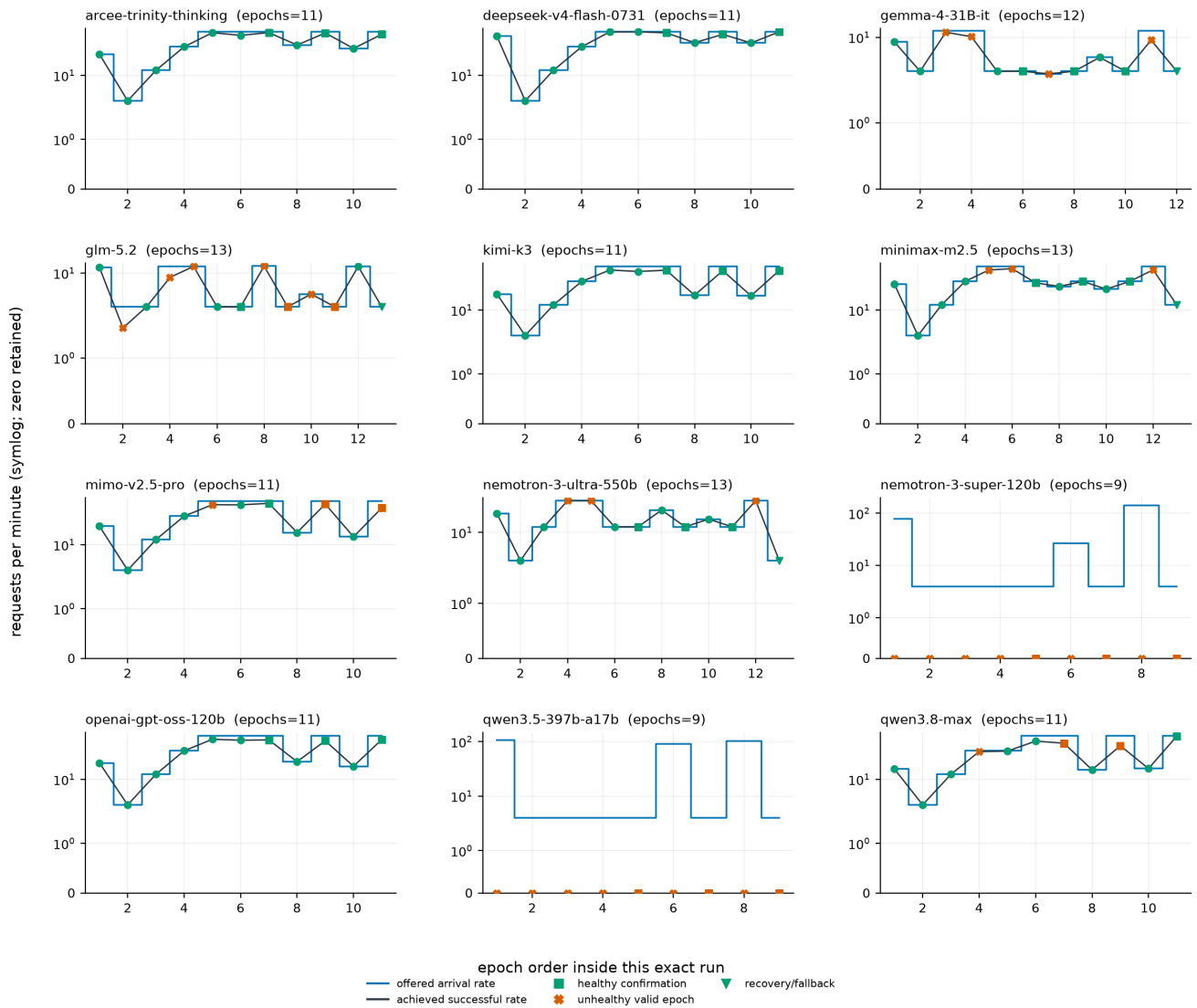
Chronological AIMD controller: short short
do-direct-aimd-20260823-final



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Input32k Short Do Sixhour AIMD 20260824 R1

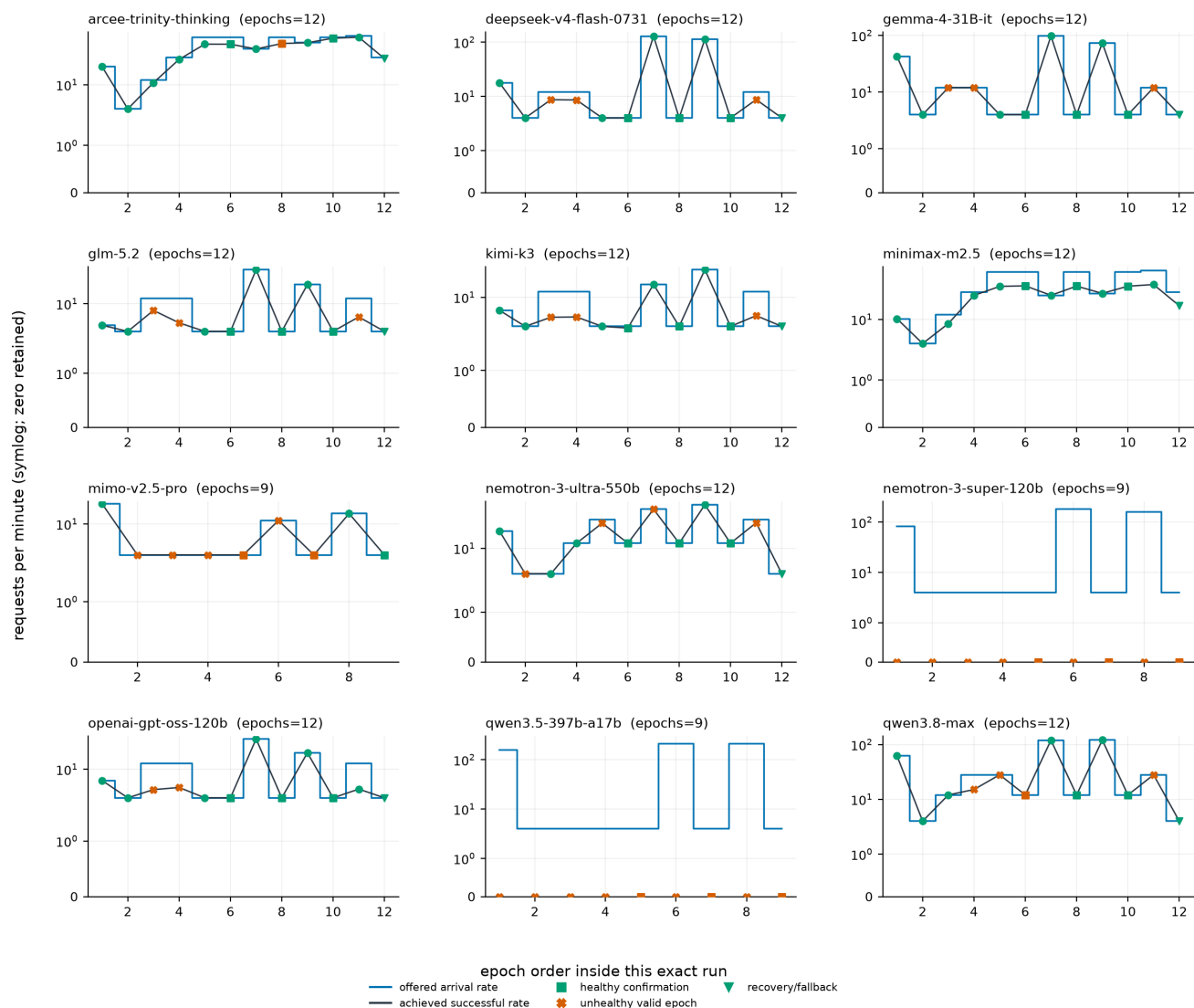
Chronological AIMD controller: input32k short
do-sixhour-aimd-20260824-r1



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Mixed Do Sixhour AIMD 20260824 R1

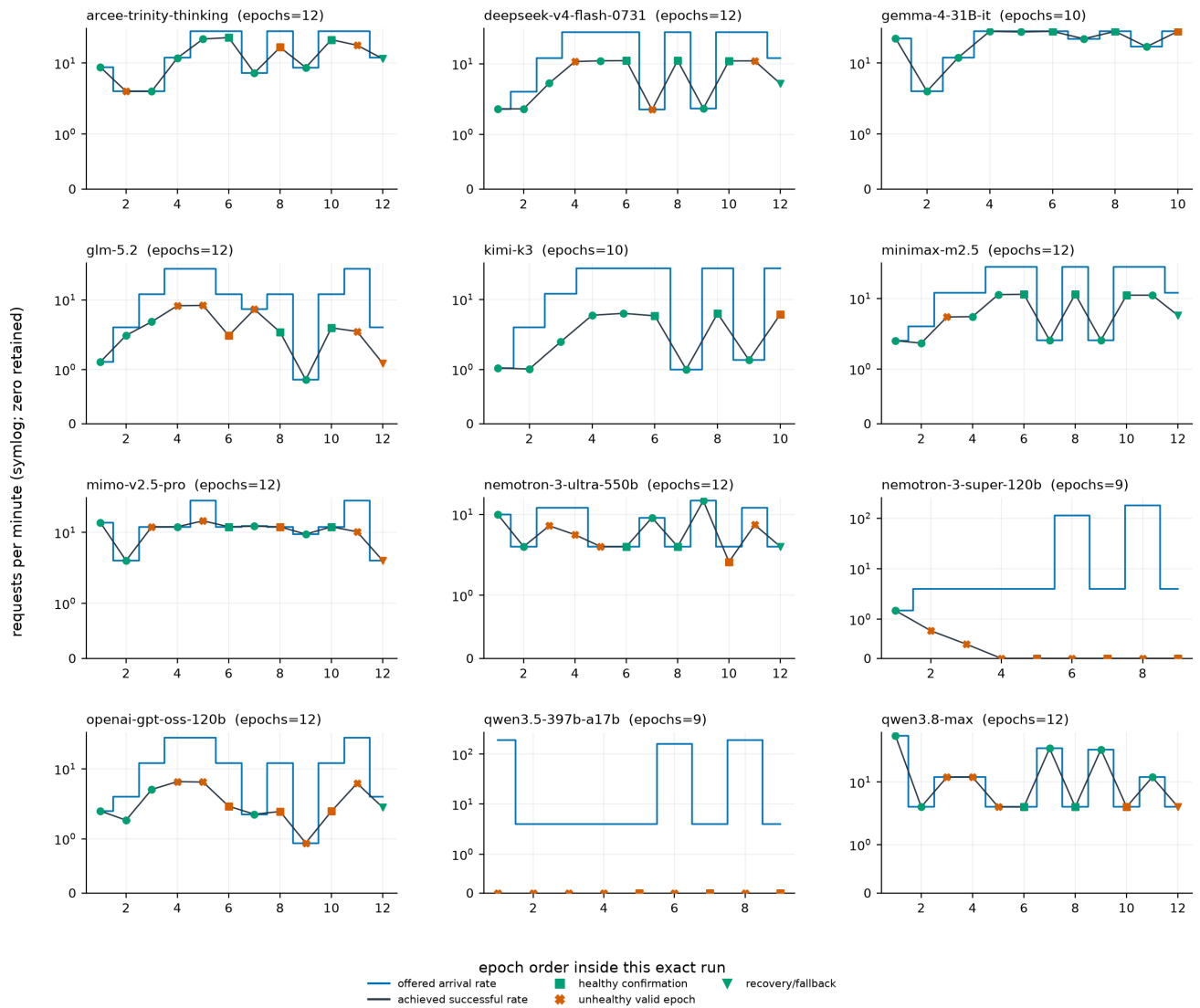
Chronological AIMD controller: mixed
do-sixhour-aimd-20260824-r1



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Short Long Do Sixhour AIMD 20260824 R1

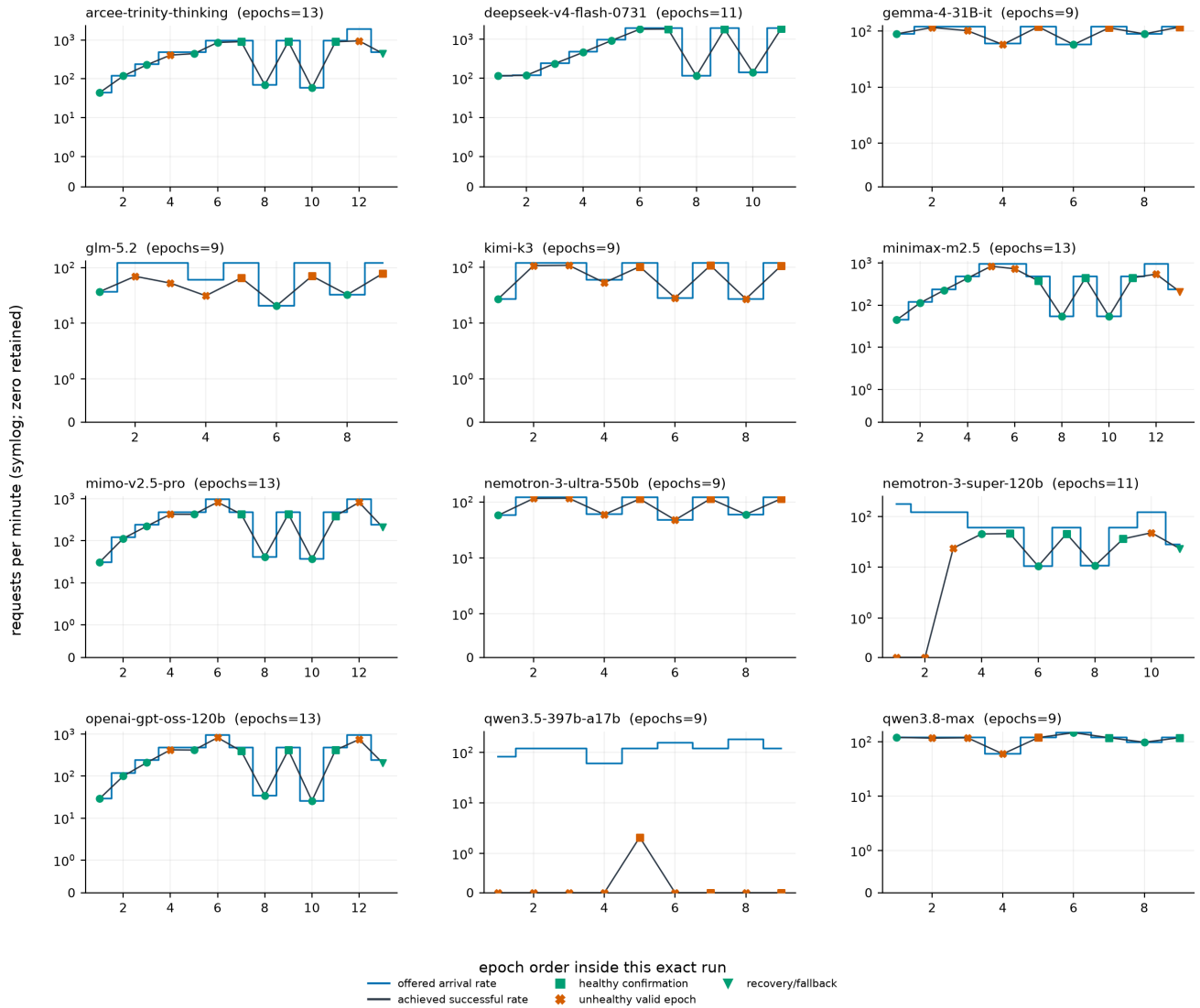
Chronological AIMD controller: short long
do-sixhour-aimd-20260824-r1



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

AIMD Controller Short Short Do Sixhour AIMD 20260824 R1

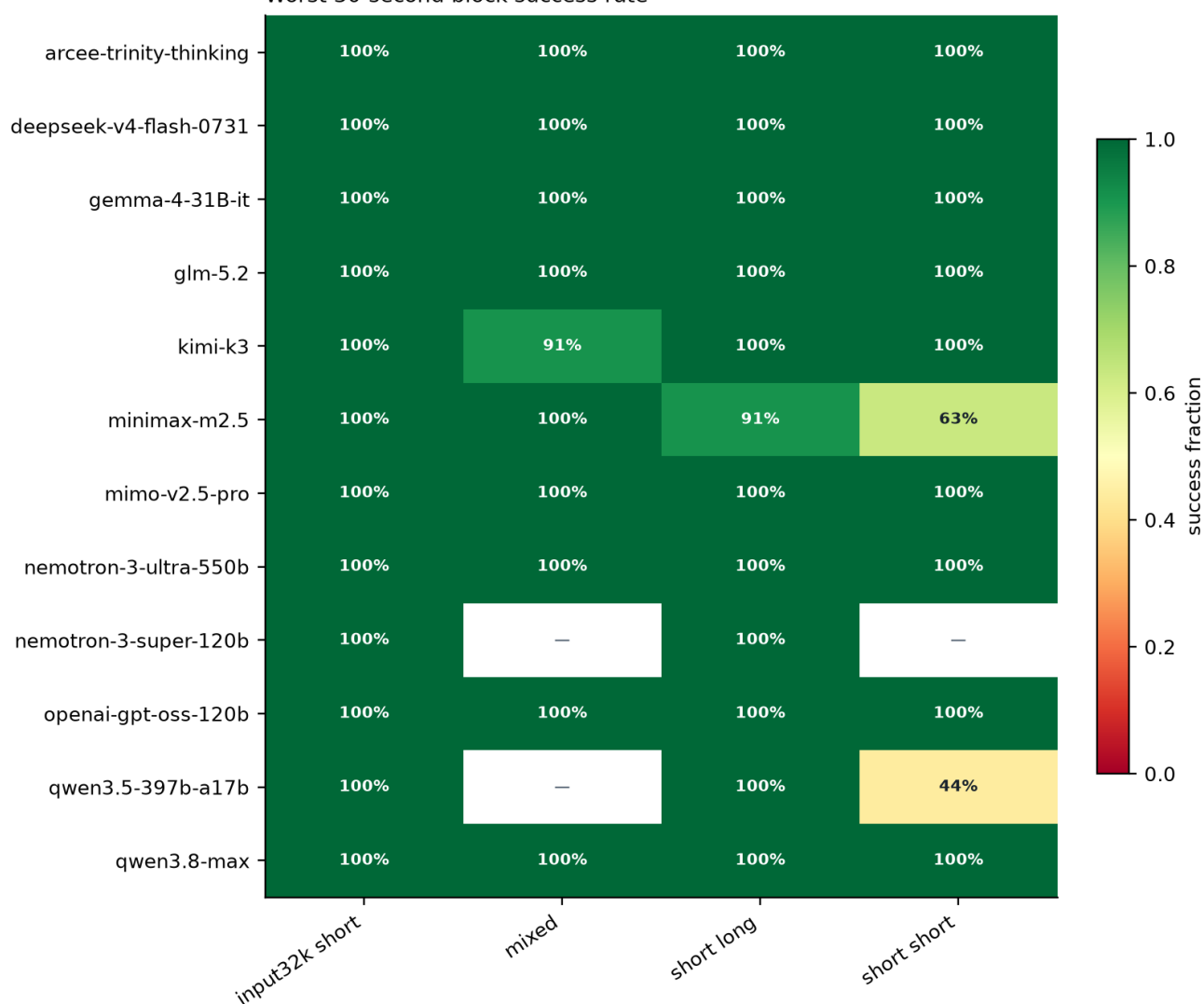
Chronological AIMD controller: short short
do-sixhour-aimd-20260824-r1



Chronology is connected only within one exact source run and workload. The blue step is offered arrival load; the dark line is drain-inclusive successful throughput. Markers identify confirmations, congestion, and recovery/fallback without connecting different endpoints or workloads.

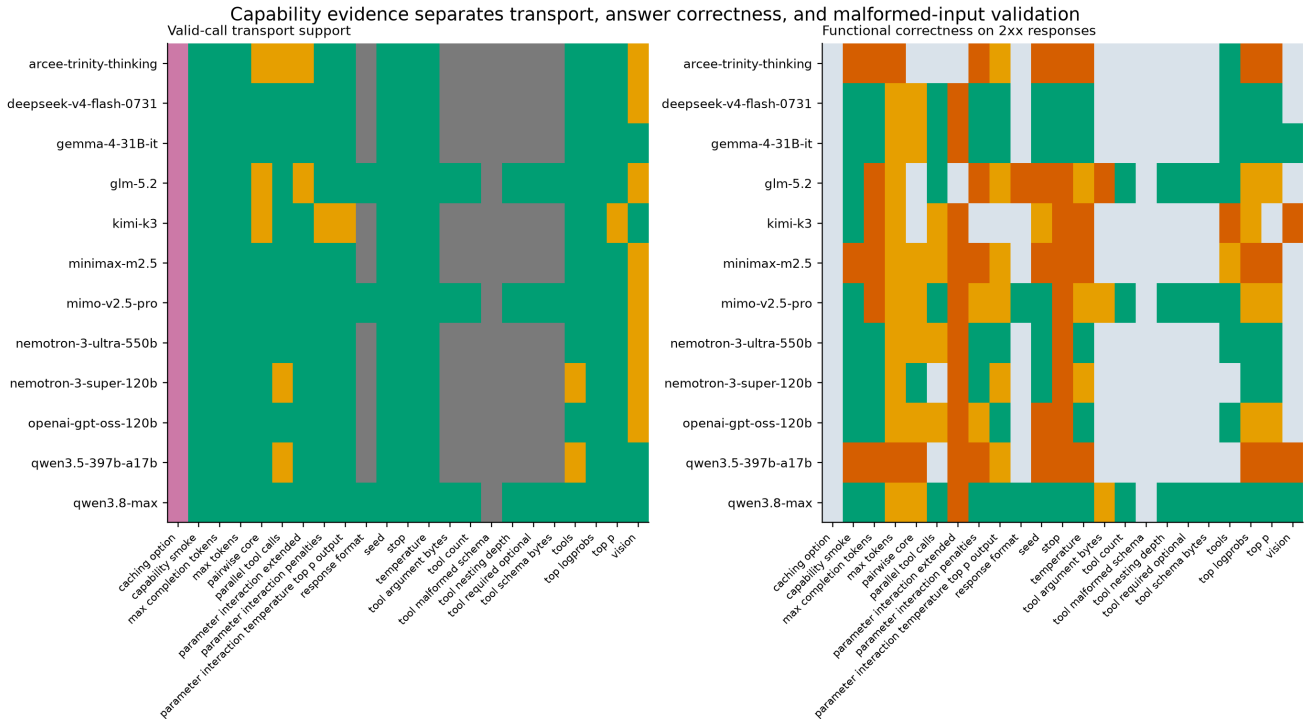
Soak Four Block Stability

Two-minute soak stability: four predeclared 30-second analysis blocks
Worst 30-second block success rate



Each cell uses four fixed 30-second analysis_block_id units. Missing cells are not zero; they mean the block estimand was unavailable.

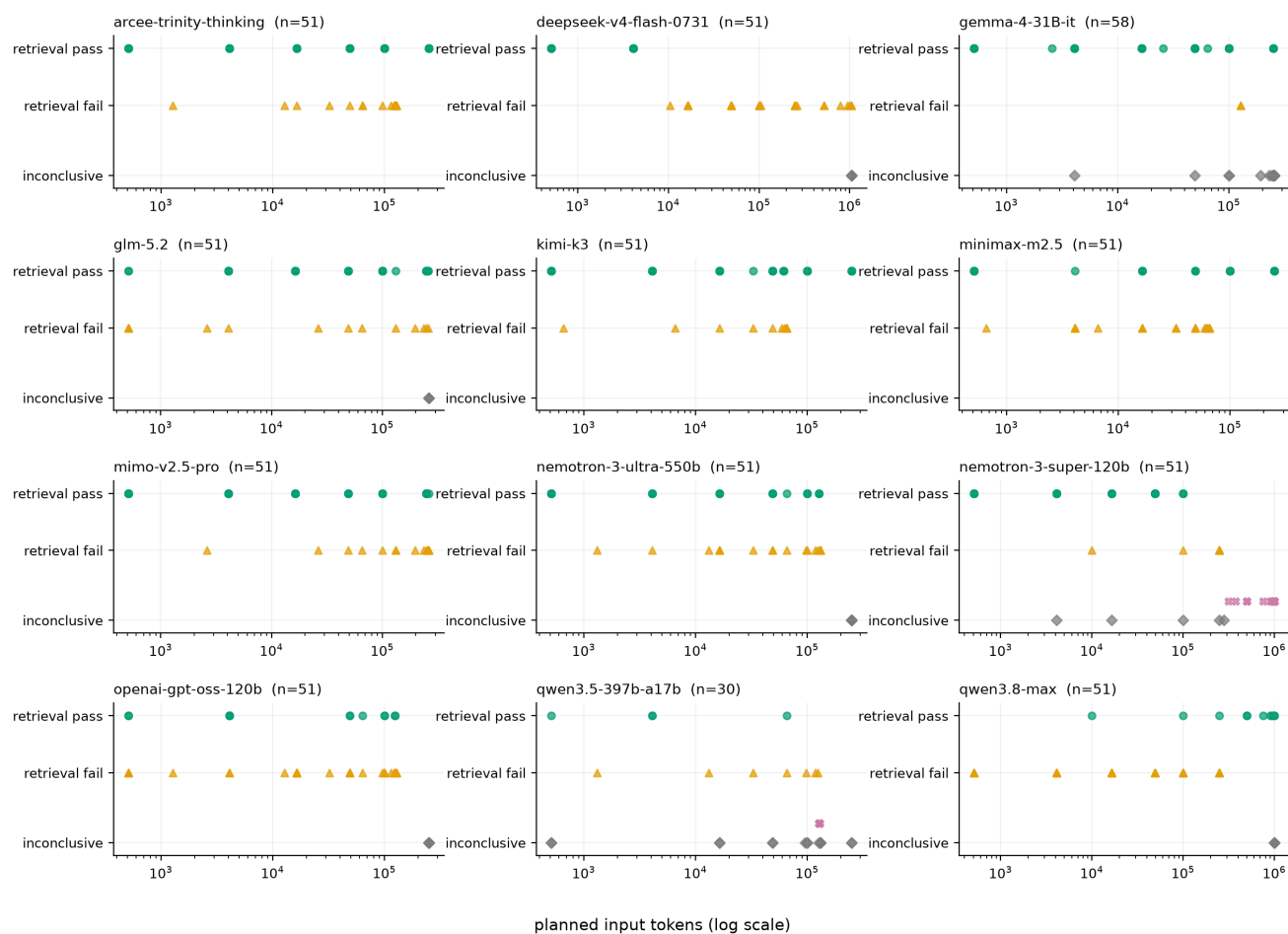
Capability Transport Functional Matrix



Valid-call transport support and task correctness are separate. Malformed-input rejection is reported in the capability evidence table and never used to infer lack of valid-call support.

Context Acceptance Retrieval

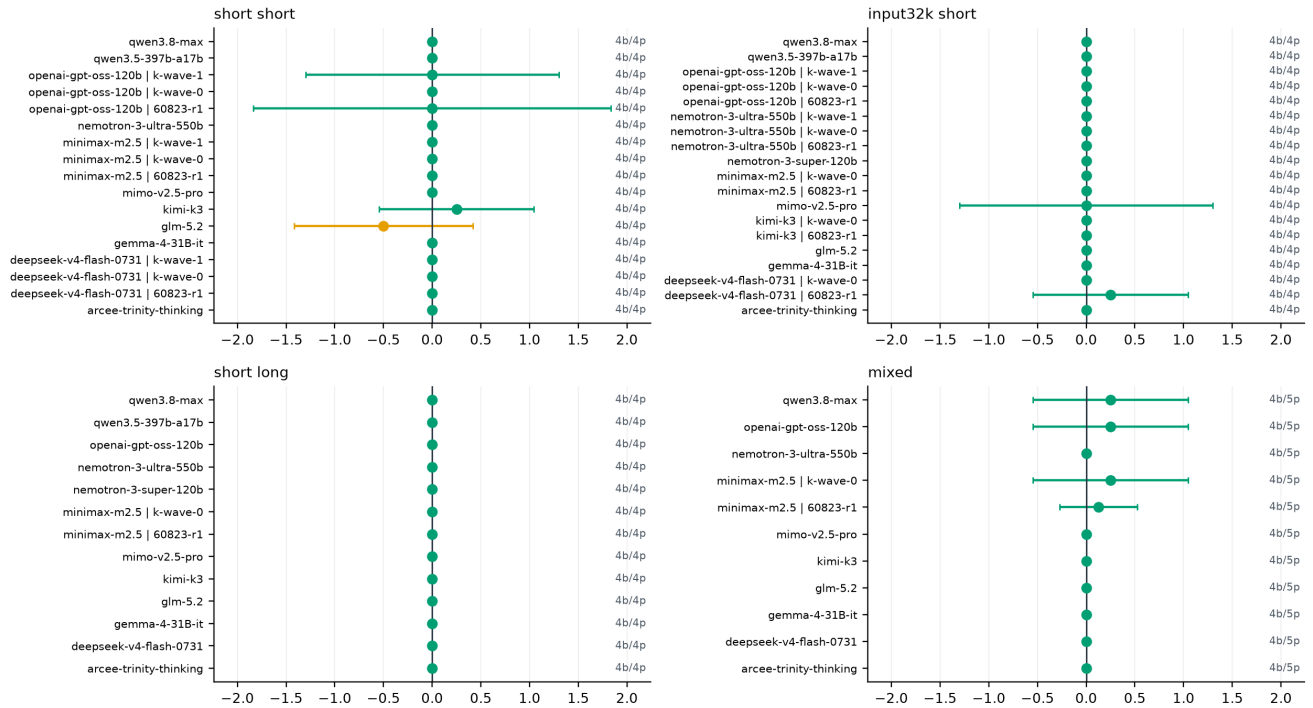
Long-context acceptance and retrieval outcome (concurrent timing is not compared)



Intervals and evidence labels come from the public analysis bundle. Blank or missing cells mean untested, censored, or inconclusive - not zero.

Paired Quality Load Effect

Paired quality change under load: exact endpoint and workload cells



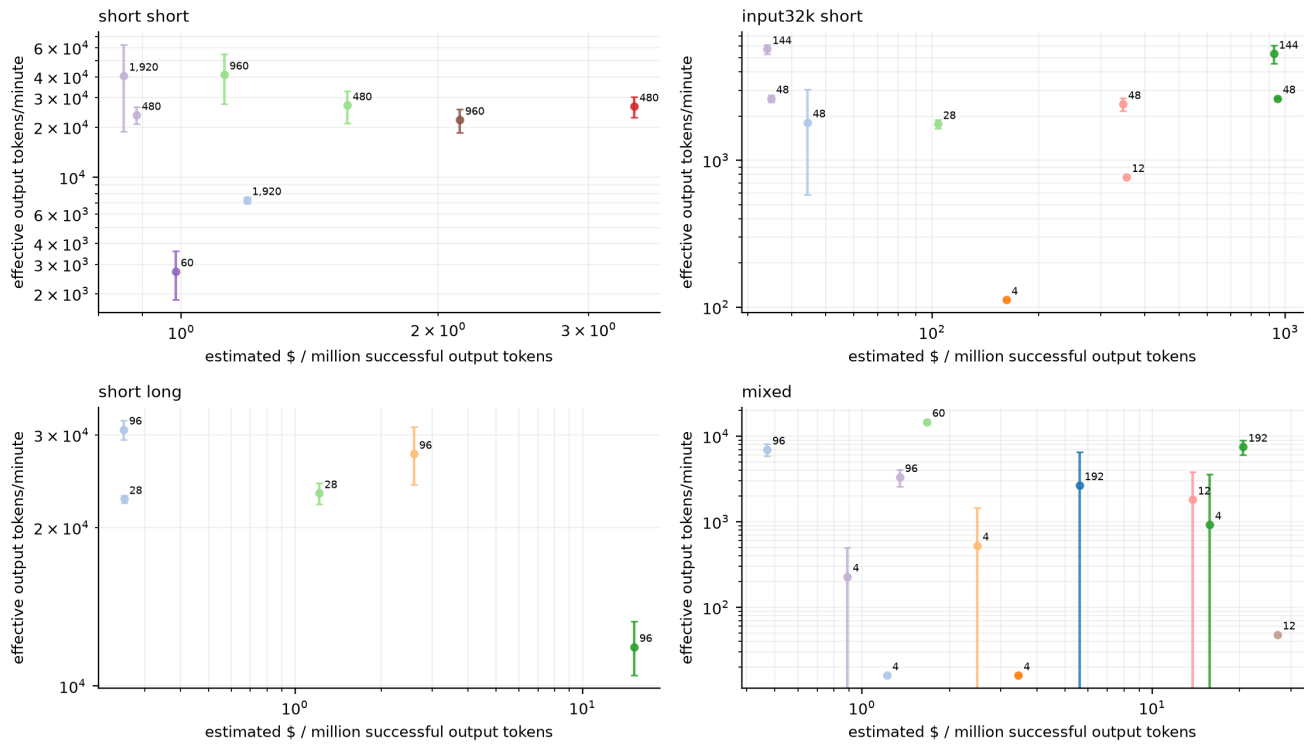
near-load minus matched low-load quality score; bars are 95% t intervals across block means

Right-edge labels report independent analysis blocks (b) and exact matched pairs (p).

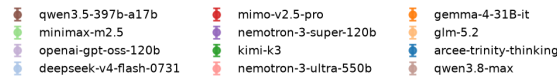
Only exact-payload matched low-load/near-load pairs enter the figure. Pair deltas are averaged within each predeclared soak analysis block; the Student-t interval uses those block means and reports both counts.

Matched Cost Performance

Exact confirmed-rate cells: cost versus achieved output goodput



Number beside each point = healthy offered RPM; bars = 95% t Interval across confirmation epochs



Every point is one exact source, endpoint, workload shape, and confirmed healthy offered rate with at least three confirmation epochs. Bars use the epoch output-TPM values. Phases, offered rates, and runs are never pooled.

5. Endpoint-by-endpoint engineering profiles

Each profile uses the same inventory, coverage, capacity, soak, limit, and quality structure so engineers can compare endpoints directly.

5.1. arcee-trinity-large-thinking

Inventory

Exact endpoint / model ID	arcee-trinity-large-thinking
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.250000
Output price per million tokens	\$0.900000
Request-attributed estimated cost (this endpoint)	\$9.49
Documented context tokens	128,000
Documented output tokens	32,000
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	9	A planned cell produced usable evidence.
Inconclusive	6	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Failed (0/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed transport degraded (0/33 2xx)	Not scored (0/0)	Not tested (n=0)	33
Pairwise core	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parallel tool calls	Observed transport degraded (0/2 2xx)	Not scored (0/0)	Not tested (n=0)	2
Parameter interaction extended	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Failed (0/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Degraded (2/6)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Tools	Observed supported (1/2 2xx)	Passed (1/1)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Top p	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	144	112.84 to 115.52	2,525,755 to 2,585,841	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	192	84.01 to 101.89	47,528 to 57,685	1,586 to 4,468
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	48	43.85 to 46.72	1,006,049 to 1,071,874	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
Short input / short output	do-sixhour-aimd-20260824-r1	Confirmed lower bound (right-censored)	960	898.18 to 903.43	41,889 to 42,352	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
32K input / short output do-direct-aimd-20260823-final	3 / 36	Not observed	Not observed	streamed	Not observed	6.23-6.38 s
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 48	21-2,855	4-568	streamed	Not observed	9.42-11.43 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 36	Not observed	Not observed	streamed	Not observed	15.41-16.42 s
Short input / short output do-sixhour-aimd-20260824-r1	3 / 720	Not observed	Not observed	streamed	Not observed	15.94-16.03 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
32K input / short output do-direct-aimd-20260823-final	36 / 36	0 / 0 / 0	1.25 (1.23 to 1.26)	2.37 (2.02 to 2.86)
Heterogeneous mixed do-direct-aimd-20260823-final	48 / 48	0 / 0 / 0	0.66 (0.64 to 0.67)	6.92 (5.4 to 8.19)
32K input / short output do-sixhour-aimd-20260824-r1	36 / 36	0 / 0 / 0	1.51 (1.44 to 1.57)	2.69 (2.31 to 3.07)
Short input / short output do-sixhour-aimd-20260824-r1	720 / 720	0 / 0 / 0	0.57 (0.57 to 0.58)	1.46 (1.45 to 1.47)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
Short input / long output	Not soak-verified	Complete; measured fail	37.32 (28.78 to 45.86)	Not reported	Not reported
Short input / short output	Not soak-verified	Complete; measured fail	59.86 (59.55 to 60.18)	1,922	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	24 (24 to 24)	537,240	Not reported
Heterogeneous mixed	Not soak-verified	Complete; measured fail	22.12 (19.42 to 24.82)	Not reported	Not reported

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
Short input / long output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	2.12 / 22.61
Short input / short output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	0.69 / 1.33
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	1.6 / 4.52
Heterogeneous mixed	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	1 / 6.79

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	128000	174988	174988	Not observed	Right censored
combined prompt + requested output	128000	132320	Not verified	Not observed	Right censored
output limit	32000	32001	847	64	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1012650040428	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1078309756652	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-186666500814	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-238995738548	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-254367155293	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-271716410559	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-367216940862	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-407913231375	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-4460434212	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-496090302304	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-531729406742	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-53462531473	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	4 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4

5.2. deepseek-v4-flash-0731

Inventory

Exact endpoint / model ID	deepseek-v4-flash-0731
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.080000
Output price per million tokens	\$0.252000
Request-attributed estimated cost (this endpoint)	\$4.30
Documented context tokens	1,048,576
Documented output tokens	1,048,576
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	8	A planned cell produced usable evidence.
Inconclusive	7	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (2/3 2xx)	Degraded (1/2)	Correct rejection observed (n=1)	4
Pairwise core	Observed supported (33/33 2xx)	Degraded (20/33)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (7/7 2xx)	Degraded (3/7)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (6/7 2xx)	Failed (0/6)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Passed (3/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Passed (6/6)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Passed (5/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	96	23.98 to 27.28	17,863 to 20,351	6,540 to 7,444
Short input / long output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	96	14.7 to 15.16	839.5 to 864.16	30,097 to 31,049
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	48	43.86 to 48	958,715 to 1,049,132	1,472 to 2,376
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	4 to 4	184 to 204	16 to 16
Short input / long output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	28	10.99 to 11.13	795.65 to 812.8	22,498 to 22,803
Short input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	1,920	1,790 to 1,858	84,455 to 87,661	7,161 to 7,433

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 24	28-2,722	4-1,024	streamed	Not observed	17.59-20.02 s
Short input / long output do-direct-aimd-20260823-final	3 / 24	57-58	2,048	streamed	Not observed	31.66-32.66 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 36	21,854-21,861	5-64	streamed	Not observed	15-16.41 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	46-51	4	streamed	Not observed	15 s
Short input / long output do-sixhour-aimd-20260824-r1	3 / 21	70-76	2,048	streamed	Not observed	37.72-38.23 s
Short input / short output do-sixhour-aimd-20260824-r1	3 / 1440	40-54	4	streamed	Not observed	15.5-16.09 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Heterogeneous mixed do-direct-aimd-20260823-final	24 / 24	0 / 0 / 0	0.45 (0.42 to 0.52)	14.22 (13.75 to 14.82)
Short input / long output do-direct-aimd-20260823-final	24 / 24	0 / 0 / 0	0.44 (0.41 to 0.47)	27.59 (27.18 to 28.08)
32K input / short output do-sixhour-aimd-20260824-r1	36 / 36	0 / 0 / 0	0.99 (0.98 to 1.02)	2.25 (1.79 to 2.71)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	0.5 (0.36 to 0.58)	0.55 (0.44 to 0.62)
Short input / long output do-sixhour-aimd-20260824-r1	21 / 21	0 / 0 / 0	0.5 (0.42 to 0.57)	27.33 (26.91 to 27.68)
Short input / short output do-sixhour-aimd-20260824-r1	1440 / 1440	0 / 0 / 0	0.41 (0.4 to 0.41)	0.72 (0.69 to 0.75)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	35.99 (35.97 to 36.02)	787,331	1,437
Short input / short output	Not soak-verified	Complete; measured fail	1,317 (1,293 to 1,340)	87,727	5,268
Short input / short output	Not soak-verified	Complete; measured fail	836.77 (598.48 to 1,075)	55,700	3,347
Heterogeneous mixed	Not soak-verified	Complete; measured fail	69.38 (65.53 to 73.22)	44,436	16,001
Short input / long output	Not soak-verified	Complete; measured fail	48.82 (46.19 to 51.45)	2,829	99,985
Short input / short output	1 RPS	Complete; pass	60 (60 to 60)	1,834	240
32K input / short output	Not soak-verified	Complete; measured fail	24 (24 to 24)	511,725	974

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.3 / 0.3	Measured fail	100.0% (70.1% to 100.0%)	1.5 / 3.87
Short input / short output	12 / 12	Measured fail	100.0% (98.9% to 100.0%)	1.14 / 1.77
Short input / short output	8 / 8	Measured fail	100.0% (98.4% to 100.0%)	1.14 / 2.13
Heterogeneous mixed	0.8 / 0.8	Measured fail	100.0% (86.2% to 100.0%)	0.88 / 13.31
Short input / long output	0.8 / 0.8	Measured fail	100.0% (86.2% to 100.0%)	0.79 / 28.98
Short input / short output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	1.37 / 1.49
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	1.05 / 3.29

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	1048576	1047013	2722	Not observed	Right censored
combined prompt + requested output	1048576	1044925	Not verified	Not observed	Right censored
output limit	1048576	4096	10020	1048577	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1008232437740	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1015465573667	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1044215442600	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1053756563607	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1057257493464	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-107485520051	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-109676469846	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1096995278016	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-115068971065	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-128437112249	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-141752020564	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-142763893661	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0.25	-0.55 to 1.05	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	4 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	4 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	4 / 4
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-1	4 / 4	0	0 to 0	4 / 4

5.3. gemma-4-31B-it

Inventory

Exact endpoint / model ID	gemma-4-31B-it
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.180000
Output price per million tokens	\$0.500000
Request-attributed estimated cost (this endpoint)	\$9.93
Documented context tokens	256,000
Documented output tokens	8,192
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	9	A planned cell produced usable evidence.
Inconclusive	7	The attempt ran, but evidence was insufficient for a claim.
Unsupported	0	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed supported (28/33 2xx)	Degraded (15/28)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (6/7 2xx)	Degraded (2/6)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Passed (3/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Passed (6/6)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Passed (5/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Vision	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	4 to 4	100,584 to 100,596	112 to 112
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	4 to 4	260 to 268	16 to 16
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
32K input / short output do-sixhour-aimd-20260824-r1	3 / 3	25,146-25,149	28	streamed	Not observed	15 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	65-67	4	streamed	Not observed	15 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
32K input / short output do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	6.84 (6.67 to 6.99)	8.76 (8.3 to 9.43)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	0.59 (0.33 to 0.83)	0.64 (0.39 to 0.86)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
Heterogeneous mixed	Not soak-verified	Complete; measured fail	23.74 (23.14 to 24.35)	Not reported	Not reported
Short input / long output	Not soak-verified	Complete; measured fail	20.85 (17.87 to 23.83)	Not reported	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	16.13 (9.31 to 22.95)	395,599	472.63
Short input / short output	1 RPS	Complete; pass	58.15 (52.91 to 63.4)	2,609	232.62

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
Heterogeneous mixed	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.98 / 4.18
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.76 / 10.37
32K input / short output	0.2 / 0.2	Pass	100.0% (61.0% to 100.0%)	4.27 / 8.89
Short input / short output	0.5 / 0.5	Pass	100.0% (79.6% to 100.0%)	1.24 / 2.3

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	256000	251873	251873	Not observed	Right censored
combined prompt + requested output	256000	130296	Not verified	Not observed	Right censored
output limit	8192	16384	947	Not observed	Realized generation observed
vision	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-262697170495	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-366009862692	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-544473296520	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-727087816062	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-99352222570	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1020877548045	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1043612387836	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-115002335213	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-126983686210	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-182984803774	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-227617460957	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-252580388130	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	4 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	4 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	4 / 4

5.4. glm-5.2

Inventory

Exact endpoint / model ID	glm-5.2
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.700000
Output price per million tokens	\$2.20
Request-attributed estimated cost (this endpoint)	\$8.12
Documented context tokens	262,144
Documented output tokens	262,144
Documented streaming	Not reported
Documented tool calling	Yes
Documented structured output	Yes
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	9	A planned cell produced usable evidence.
Inconclusive	6	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed transport degraded (0/33 2xx)	Not scored (0/0)	Not tested (n=0)	33
Pairwise core	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Parameter interaction extended	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Failed (0/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Degraded (5/6)	Not tested (n=0)	6
Response format	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Seed	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Degraded (1/7)	Not tested (n=0)	7
Tool argument bytes	Observed supported (2/4 2xx)	Failed (0/2)	Not tested (n=0)	4
Tool count	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool malformed schema	Inconclusive (0/0 2xx)	Not scored (0/0)	Invalid input accepted (n=3)	3
Tool nesting depth	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool required optional	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Tool schema bytes	Observed supported (5/5 2xx)	Passed (5/5)	Not tested (n=0)	5
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Degraded (1/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Degraded (2/7)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	96	12.92 to 22.79	852.8 to 942.99	26,463 to 29,173
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	4 to 4	220 to 224	100 to 784
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
Short input / short output	do-sixhour-aimd-20260824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Short input / long output do-direct-aimd-20260823-final	3 / 24	64-67	2,048	streamed	Not observed	21.06-37.15 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	55-56	25-196	streamed	Not observed	15 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Short input / long output do-direct-aimd-20260823-final	24 / 24	0 / 0 / 0	0.52 (0.48 to 0.57)	25.16 (19.8 to 32.57)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	0.58 (0.45 to 0.77)	2.44 (0.67 to 4.27)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	20.09 (14.38 to 25.8)	461,906	1,286
Short input / long output	Not soak-verified	Complete; measured fail	12.98 (8.05 to 17.91)	Not reported	Not reported
Heterogeneous mixed	Not soak-verified	Complete; measured fail	5.98 (5.93 to 6.04)	4,667	3,561
Short input / short output	Not soak-verified	Complete; measured fail	59.66 (59.03 to 60.29)	2,341	3,280

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	2.04 / 2.65
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.82 / 29.94
Heterogeneous mixed	0.05 / 0.03	Pass	100.0% (20.7% to 100.0%)	1.26 / 15.51
Short input / short output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	0.68 / 2.54

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	262144	265333	182823	Not observed	Right censored
combined prompt + requested output	262144	265289	265289	Not observed	Right censored
output limit	262144	262145	4096	Not observed	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	True	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	True	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-143006811582	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-267871753537	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-65838938538	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-77008106541	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-91604039851	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1008603364362	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1078364264318	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-175119189931	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-251023845050	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-305627500433	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-348718670758	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-421052831628	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	2 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	-0.5	-1.42 to 0.42	0 / 4

5.5. kimi-k3

Inventory

Exact endpoint / model ID	kimi-k3
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$2.85
Output price per million tokens	\$14.25
Request-attributed estimated cost (this endpoint)	\$56.64
Documented context tokens	Not reported
Documented output tokens	Not reported
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	Yes

Coverage

Coverage state	Cells	Meaning
Completed	8	A planned cell produced usable evidence.
Inconclusive	8	The attempt ran, but evidence was insufficient for a claim.
Unsupported	0	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed transport degraded (0/33 2xx)	Not scored (0/0)	Not tested (n=0)	33
Pairwise core	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (1/7 2xx)	Failed (0/1)	Not tested (n=0)	7
Parameter interaction penalties	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed transport degraded (0/6 2xx)	Not scored (0/0)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (3/7 2xx)	Failed (0/3)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Degraded (3/5)	Not tested (n=0)	7
Top p	Observed transport degraded (0/7 2xx)	Not scored (0/0)	Not tested (n=0)	7
Vision	Observed supported (18/18 2xx)	Failed (0/18)	Correct rejection observed (n=1)	19

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	144	78.81 to 87.93	1,602,935 to 1,788,447	5,044 to 5,628
Heterogeneous mixed	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	192	24.97 to 27.99	15,562 to 17,435	6,917 to 8,094
Short input / long output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	96	5.6 to 6.49	778.42 to 848.7	11,459 to 12,505
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	48	41.24 to 41.95	859,634 to 874,604	2,619 to 2,654
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	3.8 to 4	489.97 to 528	244 to 2,157
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
32K input / short output do-direct-aimd-20260823-final	3 / 36	20,339-20,340	45-64	streamed	Not observed	8.19-9.14 s
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 48	111-2,669	37-1,024	streamed	Not observed	34.3-38.44 s
Short input / long output do-direct-aimd-20260823-final	3 / 24	139-140	2,048	streamed	Not observed	73.9-85.79 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 36	20,844-20,851	45-64	streamed	Not observed	17.16-17.46 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	125-132	61-568	streamed	Not observed	15-15.8 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
32K input / short output do-direct-aimd-20260823-final	36 / 36	0 / 0 / 0	1.73 (1.64 to 1.78)	5.79 (5.24 to 6.08)
Heterogeneous mixed do-direct-aimd-20260823-final	48 / 48	0 / 0 / 0	0.96 (0.91 to 1.01)	31.34 (30.36 to 32.02)
Short input / long output do-direct-aimd-20260823-final	24 / 24	0 / 0 / 0	0.88 (0.8 to 0.93)	77.59 (72.18 to 85.32)
32K input / short output do-sixhour-aimd-20260824-r1	36 / 36	0 / 0 / 0	2.02 (1.91 to 2.16)	4.82 (4.62 to 5.03)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	0.87 (0.86 to 0.87)	7.18 (2.48 to 15.78)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	31.2 (28.88 to 33.52)	650,990	1,988
Short input / short output	Not soak-verified	Complete; measured fail	56 (53.07 to 58.94)	6,330	3,389
32K input / short output	Not soak-verified	Complete; measured fail	112.13 (88.16 to 136.1)	2,280,766	7,170
Heterogeneous mixed	Not soak-verified	Complete; measured fail	80.2 (67.46 to 92.94)	Not reported	Not reported
Short input / long output	Not soak-verified	Complete; measured fail	7.67 (7.56 to 7.77)	Not reported	Not reported

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.3 / 0.3	Measured fail	100.0% (70.1% to 100.0%)	2.79 / 7.87
Short input / short output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	0.66 / 3.41
32K input / short output	1.2 / 1.2	Measured fail	100.0% (90.4% to 100.0%)	1.85 / 6.17
Heterogeneous mixed	1.6 / 1.6	Measured fail	100.0% (92.6% to 100.0%)	1.32 / 36.26
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	1.39 / 69.4

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	Not reported	158411	158411	Not observed	Right censored
combined prompt + requested output	Not reported	53633	53633	Not observed	Right censored
output limit	Not reported	65537	4096	64	Realized generation observed
vision	True	observed_functional	observed_functional	Not observed	Functional in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1065717352340	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-132837199547	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-173559517454	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-241087996785	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-279437813502	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-545789265564	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-66849861044	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-827297446932	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-908766842251	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-924357251107	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-100113688088	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-106049039417	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	4 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0.25	-0.55 to 1.05	2 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4

5.6. minimax-m2.5

Inventory

Exact endpoint / model ID	minimax-m2.5
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.300000
Output price per million tokens	\$1.20
Request-attributed estimated cost (this endpoint)	\$3.98
Documented context tokens	65,536
Documented output tokens	65,536
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	6	A planned cell produced usable evidence.
Inconclusive	9	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed supported (6/33 2xx)	Degraded (1/6)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (2/7 2xx)	Degraded (1/2)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (1/7 2xx)	Failed (0/1)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Failed (0/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (4/6 2xx)	Degraded (3/4)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (6/7 2xx)	Failed (0/6)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Top p	Observed supported (4/7 2xx)	Failed (0/4)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	960	569.62 to 739.5	36,085 to 46,810	36,456 to 47,328
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed bracket	28	26.63 to 28	582,820 to 612,836	1,704 to 1,792
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	60	35.15 to 35.61	23,010 to 23,304	14,372 to 14,857
Short input / long output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	28	11.11 to 11.5	1,141 to 1,198	22,753 to 23,558
Short input / short output	do-sixhour-aimd-202 60824-r1	Confirmed bracket	480	379.19 to 446.66	29,507 to 34,896	24,268 to 28,586

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Short input / short output do-direct-aimd-20260823-final	3 / 240	59-69	64	streamed	Not observed	6.49-8.43 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 21	21,884-21,888	64	streamed	Not observed	15-15.77 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 45	76-2,767	61-1,024	streamed	Not observed	25.27-25.6 s
Short input / long output do-sixhour-aimd-20260824-r1	3 / 21	100-110	2,048	streamed	Not observed	36.51-37.8 s
Short input / short output do-sixhour-aimd-20260824-r1	3 / 360	72-84	64	streamed	Not observed	16.12-18.99 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Short input / short output do-direct-aimd-20260823-final	240 / 240	0 / 0 / 0	0.37 (0.36 to 0.37)	1.77 (1.66 to 1.92)
32K input / short output do-sixhour-aimd-20260824-r1	21 / 21	0 / 0 / 0	1.66 (1.6 to 1.74)	3.71 (3.24 to 4.31)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	45 / 45	0 / 0 / 0	0.49 (0.46 to 0.53)	14.96 (14.74 to 15.08)
Short input / long output do-sixhour-aimd-20260824-r1	21 / 21	0 / 0 / 0	0.5 (0.44 to 0.52)	26.25 (25.96 to 26.8)
Short input / short output do-sixhour-aimd-20260824-r1	360 / 360	0 / 0 / 0	0.38 (0.38 to 0.38)	2.11 (1.8 to 2.26)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	19.95 (18.15 to 21.75)	436,957	1,277
Short input / short output	Not soak-verified	Complete; measured fail	269.67 (250.25 to 289.08)	25,914	17,259
Heterogeneous mixed	Not soak-verified	Complete; measured fail	18.81 (6.7 to 30.92)	13,290	8,756
Short input / long output	Not soak-verified	Complete; measured fail	5.79 (0.79 to 10.8)	701.09	11,867
Short input / short output	Not soak-verified	Complete; measured fail	189.83 (174.07 to 205.6)	18,283	12,149
Short input / long output	Not soak-verified	Complete; measured fail	12.39 (12.23 to 12.56)	1,115	25,381
32K input / short output	Not soak-verified	Complete; measured fail	24 (24 to 24)	512,447	1,536
Heterogeneous mixed	Not soak-verified	Complete; measured fail	10.75 (8.37 to 13.13)	7,202	4,623

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.17 / 0.17	Measured fail	100.0% (56.6% to 100.0%)	2.52 / 6.95
Short input / short output	3 / 3	Measured fail	100.0% (95.9% to 100.0%)	1.09 / 7.39
Heterogeneous mixed	0.38 / 0.37	Measured fail	100.0% (74.1% to 100.0%)	1.02 / 30.58
Short input / long output	0.17 / 0.17	Measured fail	100.0% (56.6% to 100.0%)	1.41 / 116.26
Short input / short output	2 / 2	Measured fail	100.0% (94.0% to 100.0%)	1.12 / 7.97
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.6 / 30.72
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	2.34 / 7.6
Heterogeneous mixed	0.1 / 0.1	Measured fail	100.0% (43.9% to 100.0%)	0.45 / 12.72
Short input / short output	8 / 8	Measured fail	100.0% (98.4% to 100.0%)	0.57 / 2.03

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	65536	166688	166688	Not observed	Right censored
combined prompt + requested output	65536	60051	Not verified	Not observed	Right censored
output limit	65536	65537	4096	64	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1010742273091	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1018337971843	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1024762958607	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1025488084190	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-102973946283	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1039653108785	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1041420121351	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1047787868800	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1055732441835	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1057054081827	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1064296565114	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1091222788197	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0.12	-0.27 to 0.52	2 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-sixhour-completion-20260824-r1-soak-wave-0	4 / 5	0.25	-0.55 to 1.05	3 / 5
Short input / long output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-1	4 / 4	0	0 to 0	0 / 4

5.7. mimo-v2.5-pro

Inventory

Exact endpoint / model ID	mimo-v2.5-pro
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.400000
Output price per million tokens	\$1.50
Request-attributed estimated cost (this endpoint)	\$6.14
Documented context tokens	262,144
Documented output tokens	262,144
Documented streaming	Not reported
Documented tool calling	Yes
Documented structured output	Yes
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	7	A planned cell produced usable evidence.
Inconclusive	8	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (2/3 2xx)	Degraded (1/2)	Correct rejection observed (n=1)	4
Pairwise core	Observed supported (30/33 2xx)	Degraded (17/30)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (7/7 2xx)	Degraded (4/7)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Degraded (2/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Degraded (4/6)	Not tested (n=0)	6
Response format	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Degraded (3/7)	Not tested (n=0)	7
Tool argument bytes	Observed supported (4/4 2xx)	Degraded (1/4)	Not tested (n=0)	4
Tool count	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool malformed schema	Inconclusive (0/0 2xx)	Not scored (0/0)	Invalid input accepted (n=3)	3
Tool nesting depth	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool required optional	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Tool schema bytes	Observed supported (5/5 2xx)	Passed (5/5)	Not tested (n=0)	5
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Degraded (4/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Degraded (4/7)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	480	388.95 to 431.12	117,725 to 130,615	24,870 to 27,592

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Short input / short output do-sixhour-aimd-20260824-r1	3 / 360	296-308	57-64	streamed	Not observed	16.7-18.51 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Short input / short output do-sixhour-aimd-20260824-r1	360 / 360	0 / 0 / 0	0.43 (0.41 to 0.45)	2.96 (2.88 to 3.06)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
Short input / short output	Not soak-verified	Baseline transport gate failed; inconclusive	Not reported (Exploratory / CI unavailable)	Not reported	Not reported
Short input / short output	Not soak-verified	Baseline transport gate failed; inconclusive	Not reported (Exploratory / CI unavailable)	Not reported	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	22.89 (20.67 to 25.11)	591,191	1,379
Short input / long output	Not soak-verified	Complete; measured fail	9.99 (5.59 to 14.39)	Not reported	Not reported
Heterogeneous mixed	Not soak-verified	Complete; measured fail	8.67 (7.17 to 10.16)	Not reported	Not reported
Short input / short output	Not soak-verified	Complete; measured fail	24 (24 to 24)	6,782	1,536

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	6.15 / 9.74
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.5 / 72.8
Heterogeneous mixed	0.1 / 0.1	Measured fail	100.0% (43.9% to 100.0%)	0.68 / 7
Short input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.62 / 2.36

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	262144	246006	245518	Not observed	Right censored
combined prompt + requested output	262144	246272	Not verified	Not observed	Right censored
output limit	262144	16384	4096	262145	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	True	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	True	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-384502785565	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-43223811143	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-4802247582	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-808510244289	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-8140037340	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-103494738851	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1083859928845	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-271599621668	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-274846795325	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-313906033826	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-331384056880	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-414881359334	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	-1.3 to 1.3	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	3 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4

5.8. nemotron-3-ultra-550b

Inventory

Exact endpoint / model ID	nemotron-3-ultra-550b
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.900000
Output price per million tokens	\$1.70
Request-attributed estimated cost (this endpoint)	\$9.50
Documented context tokens	131,072
Documented output tokens	131,072
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	10	A planned cell produced usable evidence.
Inconclusive	5	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (2/3 2xx)	Degraded (1/2)	Correct rejection observed (n=1)	4
Pairwise core	Observed supported (32/33 2xx)	Degraded (19/32)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (7/7 2xx)	Degraded (1/7)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Passed (3/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Passed (6/6)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Passed (5/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Confirmed bracket	48	36.29 to 39.31	890,282 to 964,392	2,322 to 2,516
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed bracket	12	12 to 12	301,864 to 301,880	768 to 768
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	12	12 to 12	12,944 to 12,956	1,012 to 2,616
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
32K input / short output do-direct-aimd-20260823-final	3 / 12	24,534-24,537	64	streamed	Not observed	6.11-6.61 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 9	25,154-25,158	64	streamed	Not observed	15 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 9	70-3,167	52-600	streamed	Not observed	15 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
32K input / short output do-direct-aimd-20260823-final	12 / 12	0 / 0 / 0	1.79 (1.65 to 1.95)	4.8 (4.6 to 4.94)
32K input / short output do-sixhour-aimd-20260824-r1	9 / 9	0 / 0 / 0	2.23 (2.2 to 2.25)	2.98 (2.92 to 3.05)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	9 / 9	0 / 0 / 0	0.6 (0.51 to 0.67)	5.6 (3.36 to 7.24)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	8.29 (7.16 to 9.42)	208,735	530.58
32K input / short output	Not soak-verified	Complete; measured fail	6 (6 to 6)	151,056	384
Short input / long output	Not soak-verified	Complete; measured fail	24.51 (21.39 to 27.64)	Not reported	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	44.56 (40.84 to 48.28)	1,093,371	2,852
Short input / short output	Not soak-verified	Complete; measured fail	59.94 (59.81 to 60.07)	2,898	3,609
Heterogeneous mixed	Not soak-verified	Complete; measured fail	19.87 (15.48 to 24.26)	Not reported	Not reported

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.08 / 0.07	Measured fail	100.0% (34.2% to 100.0%)	2.85 / 4.19
32K input / short output	0.05 / 0.03	Measured fail	100.0% (20.7% to 100.0%)	2.22 / 3.32
Short input / long output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	0.82 / 29.8
32K input / short output	0.4 / 0.4	Measured fail	100.0% (75.8% to 100.0%)	2.79 / 3.98
Short input / short output	0.5 / 0.5	Measured fail	100.0% (79.6% to 100.0%)	1.04 / 3.72
Heterogeneous mixed	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.8 / 9.82

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	131072	126009	76667	Not observed	Right censored
combined prompt + requested output	131072	126176	126176	Not observed	Right censored
output limit	131072	16384	4096	131073	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-195412486978	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-218140183247	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-29602675950	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-732670324091	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-811208222623	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1033720579907	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1056294599172	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1066028549574	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1073885501182	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1095588156617	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-130428875085	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-162289009854	do-direct-aimd-20260823-final Confirmation	1	100.0%	20.7% to 100.0%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0	0 to 0	3 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	2 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-1	4 / 4	0	0 to 0	0 / 4

5.9. nvidia-nemotron-3-super-120b

Inventory

Exact endpoint / model ID	nvidia-nemotron-3-super-120b
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.300000
Output price per million tokens	\$0.650000
Request-attributed estimated cost (this endpoint)	\$29.64
Documented context tokens	1,000,000
Documented output tokens	32,768
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	5	A planned cell produced usable evidence.
Inconclusive	10	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed supported (3/33 2xx)	Degraded (2/3)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (1/7 2xx)	Passed (1/1)	Not tested (n=0)	7
Parallel tool calls	Observed transport degraded (0/2 2xx)	Not scored (0/0)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (1/7 2xx)	Failed (0/1)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Passed (3/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (4/6 2xx)	Degraded (3/4)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (5/7 2xx)	Degraded (2/5)	Not tested (n=0)	7
Tools	Observed transport degraded (0/2 2xx)	Not scored (0/0)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Passed (5/5)	Not tested (n=0)	7
Top p	Observed supported (4/7 2xx)	Passed (4/4)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	12	0.9 to 4.87	75.11 to 75.11	1,853 to 1,853
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Confirmed bracket	60	36.07 to 45.75	2,570 to 3,267	2,308 to 2,928

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Short input / long output do-direct-aimd-20260823-final	3 / 3	83	2,048	streamed	Not observed	12.32-66.3 s
Short input / short output do-sixhour-aimd-20260824-r1	3 / 45	68-75	64	streamed	Not observed	19.67-24.95 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Short input / long output do-direct-aimd-20260823-final	3 / 3	0 / 0 / 0	0.98 (0.43 to 1.79)	33.35 (12.32 to 66.29)
Short input / short output do-sixhour-aimd-20260824-r1	45 / 45	0 / 0 / 0	0.57 (0.56 to 0.58)	6.93 (5.99 to 7.83)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
Short input / short output	Not soak-verified	Baseline transport gate failed; inconclusive	Not reported (Exploratory / CI unavailable)	Not reported	Not reported
Short input / long output	Not soak-verified	Complete; measured fail	1.54 (-0.28 to 3.36)	Not reported	Not reported
Heterogeneous mixed	Not soak-verified	Baseline transport gate failed; inconclusive	Not reported (Exploratory / CI unavailable)	Not reported	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	22.87 (22.57 to 23.17)	561,176	1,464

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
Short input / long output	0.05 / 0.03	Measured fail	100.0% (20.7% to 100.0%)	0.63 / 30.3
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	1.23 / 4.05

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	1000000	248693	76667	312500	Interval censored
combined prompt + requested output	1000000	Not observed	Not verified	504096	Left censored
output limit	32768	32769	4096	64	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	inconclusive	inconclusive	Not observed	Inconclusive
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-214256704974	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-303838719906	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-323893650735	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-401633039697	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-423863633458	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-602581367417	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-880885659400	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-898086355679	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-931378165656	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-951953358068	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1048951846029	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1077296104995	do-direct-aimd-20260823-final Confirmation	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4

5.10. openai-gpt-oss-120b

Inventory

Exact endpoint / model ID	openai-gpt-oss-120b
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.100000
Output price per million tokens	\$0.700000
Request-attributed estimated cost (this endpoint)	\$2.58
Documented context tokens	128,000
Documented output tokens	4,096
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	10	A planned cell produced usable evidence.
Inconclusive	5	The attempt ran, but evidence was insufficient for a claim.
Unsupported	1	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed supported (22/33 2xx)	Degraded (12/22)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (5/7 2xx)	Degraded (2/5)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Degraded (1/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (4/7 2xx)	Failed (0/4)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Degraded (2/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Passed (6/6)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Degraded (4/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Degraded (5/7)	Not tested (n=0)	7
Vision	Observed transport degraded (0/1 2xx)	Not scored (0/0)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	144	86.67 to 91.81	1,852,949 to 1,962,926	5,547 to 5,876
Heterogeneous mixed	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	96	11.78 to 12.74	8,996 to 9,731	2,984 to 3,533
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Confirmed lower bound (right-censored)	1,920	548.02 to 861.28	50,531 to 79,652	33,022 to 50,245
32K input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	48	40.15 to 42.01	879,796 to 920,514	2,569 to 2,688
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Confirmed bracket	4	4 to 4	428 to 440	124 to 340
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Confirmed lower bound (right-censored)	480	397.18 to 419.01	42,395 to 44,764	22,374 to 24,514

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
32K input / short output do-direct-aimd-20260823-final	3 / 36	21,380-21,381	64	streamed	Not observed	7.84-8.31 s
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 24	92-2,780	38-1,024	streamed	Not observed	37.67-40.76 s
Short input / short output do-direct-aimd-20260823-final	3 / 480	88-97	25-64	streamed	Not observed	11.15-17.52 s
32K input / short output do-sixhour-aimd-20260824-r1	3 / 36	21,911-21,917	64	streamed	Not observed	17.14-17.93 s
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	107-110	31-85	streamed	Not observed	15 s
Short input / short output do-sixhour-aimd-20260824-r1	3 / 360	100-114	23-64	streamed	Not observed	17.18-18.13 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
32K input / short output do-direct-aimd-20260823-final	36 / 36	0 / 0 / 0	1.45 (1.31 to 1.54)	5.76 (5.69 to 5.82)
Heterogeneous mixed do-direct-aimd-20260823-final	24 / 24	0 / 0 / 0	0.58 (0.52 to 0.61)	35.36 (32.79 to 36.85)
Short input / short output do-direct-aimd-20260823-final	480 / 480	0 / 0 / 0	0.55 (0.54 to 0.56)	3.12 (3.02 to 3.18)
32K input / short output do-sixhour-aimd-20260824-r1	36 / 36	0 / 0 / 0	1.69 (1.57 to 1.89)	4.66 (4.41 to 4.94)
Heterogeneous mixed do-sixhour-aimd-20260824-r1	3 / 3	0 / 0 / 0	0.57 (0.45 to 0.7)	2.13 (1.48 to 2.54)
Short input / short output do-sixhour-aimd-20260824-r1	360 / 360	0 / 0 / 0	0.47 (0.45 to 0.5)	2.67 (2.59 to 2.78)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	31.69 (27.52 to 35.86)	695,045	2,028
Short input / short output	Not soak-verified	Complete; measured fail	294.8 (251.57 to 338.04)	36,856	15,790
32K input / short output	Not soak-verified	Complete; measured fail	23.49 (23.16 to 23.81)	515,082	1,503
Short input / short output	Not soak-verified	Complete; measured fail	213.4 (189.05 to 237.75)	26,681	11,437
Short input / short output	Not soak-verified	Complete; measured fail	692.23 (12.75 to 1,372)	63,948	40,334
32K input / short output	Not soak-verified	Complete; measured fail	129.06 (121.46 to 136.66)	2,759,446	8,260
Short input / long output	Not soak-verified	Complete; measured fail	10.56 (7.4 to 13.73)	Not reported	Not reported
Heterogeneous mixed	Not soak-verified	Complete; measured fail	15.91 (10.27 to 21.56)	Not reported	Not reported

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.3 / 0.3	Measured fail	100.0% (70.1% to 100.0%)	3.12 / 10.55
Short input / short output	3 / 3	Measured fail	100.0% (95.9% to 100.0%)	1.04 / 6.07
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	2.4 / 4.15
Short input / short output	2 / 2	Measured fail	100.0% (94.0% to 100.0%)	1.3 / 3.33
Short input / short output	16 / 16	Measured fail	100.0% (99.2% to 100.0%)	0.84 / 3.73
32K input / short output	1.2 / 1.2	Measured fail	100.0% (90.4% to 100.0%)	2.01 / 3.8
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.74 / 45.42
Heterogeneous mixed	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.73 / 12.1

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	128000	117477	66717	Not observed	Right censored
combined prompt + requested output	128000	117823	117823	Not observed	Right censored
output limit	4096	16384	4096	64	Realized generation observed
vision	Not reported	observed_rejected_or_unsupported	observed_rejected_or_unsupported	Not observed	Rejected or unsupported in observed tests
tool calling	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1005318848474	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1011894653064	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1018654372253	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1029598110431	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-103201846877	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1042232360507	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1064950106535	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1065095713444	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1070126661979	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-108094801648	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-1081442598909	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1090491409650	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0.25	-0.55 to 1.05	3 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	-1.84 to 1.84	0 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	0 / 4
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-0	4 / 4	0	0 to 0	2 / 4
32K input / short output do-sixhour-completion-20260824-r1-soak-wave-1	4 / 4	0	0 to 0	0 / 4

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
Short input / short output do-sixhour-completion-20260824-r1-soak-wave-1	4 / 4	0	-1.3 to 1.3	1 / 4

5.11. qwen3.5-397b-a17b

Inventory

Exact endpoint / model ID	qwen3.5-397b-a17b
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$0.550000
Output price per million tokens	\$3.50
Request-attributed estimated cost (this endpoint)	\$11.53
Documented context tokens	131,072
Documented output tokens	131,072
Documented streaming	Not reported
Documented tool calling	Not reported
Documented structured output	Not reported
Documented vision input	No

Coverage

Coverage state	Cells	Meaning
Completed	6	A planned cell produced usable evidence.
Inconclusive	10	The attempt ran, but evidence was insufficient for a claim.
Unsupported	0	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Max tokens	Observed supported (2/3 2xx)	Failed (0/2)	Correct rejection observed (n=1)	4
Pairwise core	Observed supported (3/33 2xx)	Degraded (1/3)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (1/7 2xx)	Failed (0/1)	Not tested (n=0)	7
Parallel tool calls	Observed transport degraded (0/2 2xx)	Not scored (0/0)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (1/7 2xx)	Failed (0/1)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Failed (0/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (4/6 2xx)	Degraded (3/4)	Not tested (n=0)	6
Seed	Observed supported (2/2 2xx)	Failed (0/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1
Temperature	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Tools	Observed transport degraded (0/2 2xx)	Not scored (0/0)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Top p	Observed supported (4/7 2xx)	Failed (0/4)	Not tested (n=0)	7
Vision	Observed supported (1/1 2xx)	Failed (0/1)	Not tested (n=0)	1

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Confirmed bracket	960	320.7 to 364.62	13,249 to 15,105	20,525 to 23,336
32K input / short output	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Censored no valid healthy epoch	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Short input / short output do-direct-aimd-20260823-final	3 / 240	38-44	64	streamed	Not observed	13.16-14.97 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Short input / short output do-direct-aimd-20260823-final	240 / 240	0 / 0 / 0	0.65 (0.63 to 0.68)	8.26 (7.99 to 8.59)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
32K input / short output	Not soak-verified	Complete; measured fail	19.81 (18.38 to 21.24)	506,883	1,268
Heterogeneous mixed	Not soak-verified	Baseline transport gate failed; inconclusive	Not reported (Exploratory / CI unavailable)	Not reported	Not reported
Short input / long output	Not soak-verified	Complete; measured fail	2.64 (2.57 to 2.7)	Not reported	Not reported
Short input / short output	Not soak-verified	Complete; measured fail	396.69 (317.93 to 475.45)	16,443	25,388

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
32K input / short output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	6.77 / 11.85
Short input / long output	0.1 / 0.1	Measured fail	100.0% (43.9% to 100.0%)	0.79 / 105.33
Short input / short output	8 / 8	Measured fail	100.0% (98.4% to 100.0%)	0.8 / 6.17

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	131072	129818	98141	129761	Nonmonotonic inconclusive
combined prompt + requested output	131072	70999	70999	130810	Interval censored
output limit	131072	1024	5632	64	Realized generation observed
vision	Not reported	inconclusive	inconclusive	Not observed	Inconclusive
tool calling	Not reported	inconclusive	inconclusive	Not observed	Inconclusive
structured output	Not reported	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-100114707469	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1004506156896	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1011980799555	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1026212945362	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1026682030928	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1033385520053	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1034036971514	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1054406529248	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1076395058812	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1081729496550	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1082889354322	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%
Short input / short output direct-short-1087846670508	do-direct-aimd-20260823-final Additive aimd	1	0.0%	0.0% to 79.3%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4

5.12. qwen3.8-max

Inventory

Exact endpoint / model ID	qwen3.8-max
Display name (if exposed)	Not reported
Version / revision (if exposed)	Not reported
Serving region (if exposed)	Not reported
API version / surface	v1
Input price per million tokens	\$2.00
Output price per million tokens	\$6.00
Request-attributed estimated cost (this endpoint)	\$37.30
Documented context tokens	1,000,000
Documented output tokens	262,144
Documented streaming	Not reported
Documented tool calling	Yes
Documented structured output	Yes
Documented vision input	Yes

Coverage

Coverage state	Cells	Meaning
Completed	10	A planned cell produced usable evidence.
Inconclusive	6	The attempt ran, but evidence was insufficient for a claim.
Unsupported	0	The API or endpoint rejected the capability as unsupported.
Untested	0	No result exists; this is not a pass or a zero.

Explicitly untested capability subtests

The following manifest exclusions are untested for this endpoint. The broad endpoint-by-dimension matrix may be complete from separate evidence, but that never upgrades these named zero-attempt subtests.

Explicitly untested measurement	Coverage dimension	State	Reason
Adaptive tool +1 boundary discovery	tool_calling	Untested	not part of this fixed plan; report as untested rather than inferred
Conditional HTTP 429/5xx retry-backoff behavior	post_overload_recovery	Untested	a separate recovery experiment, not part of this no-retry capability measurement lane

Capability outcome

Transport support answers whether valid calls were served. Functional correctness scores the returned task result. Malformed validation asks whether deliberately invalid calls were rejected correctly. A generic 4xx on a malformed probe is never used to claim that valid calls are unsupported.

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Caching option	Documented unavailable (0/0 2xx)	Not scored (0/0)	Not tested (n=0)	0
Capability smoke	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max completion tokens	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Max tokens	Observed supported (3/3 2xx)	Degraded (2/3)	Invalid input accepted (n=1)	4
Pairwise core	Observed supported (31/33 2xx)	Degraded (19/31)	Not tested (n=0)	33

Capability dimension	Valid-call transport	Functional result	Malformed validation	n
Pairwise core	Observed supported (7/7 2xx)	Degraded (3/7)	Not tested (n=0)	7
Parallel tool calls	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Parameter interaction extended	Observed supported (5/7 2xx)	Failed (0/5)	Not tested (n=0)	7
Parameter interaction penalties	Observed supported (3/7 2xx)	Passed (3/3)	Not tested (n=0)	7
Parameter interaction temperature top p output	Observed supported (6/6 2xx)	Passed (6/6)	Not tested (n=0)	6
Response format	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Seed	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Stop	Observed supported (1/1 2xx)	Passed (1/1)	Not tested (n=0)	1
Temperature	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Tool argument bytes	Observed supported (2/4 2xx)	Degraded (1/2)	Not tested (n=0)	4
Tool count	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool malformed schema	Inconclusive (0/0 2xx)	Not scored (0/0)	Invalid input accepted (n=3)	3
Tool nesting depth	Observed supported (4/4 2xx)	Passed (4/4)	Not tested (n=0)	4
Tool required optional	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Tool schema bytes	Observed supported (5/5 2xx)	Passed (5/5)	Not tested (n=0)	5
Tools	Observed supported (2/2 2xx)	Passed (2/2)	Not tested (n=0)	2
Top logprobs	Observed supported (5/7 2xx)	Passed (5/5)	Not tested (n=0)	7
Top p	Observed supported (7/7 2xx)	Passed (7/7)	Not tested (n=0)	7
Vision	Observed supported (18/18 2xx)	Passed (18/18)	Correct rejection observed (n=1)	19

Capacity and latency by workload

A confirmed healthy offered rate requires three explicit healthy confirmation epochs at the same arrival rate. It describes the load sent, not the rate completed. The achieved-RPM and TPM intervals use drain-inclusive epoch outcomes. Short AIMD confirmations are never relabelled as sustained capacity; that requires the separate drain-inclusive soak evidence.

Workload	Run / source	Evidence	Confirmed healthy offered RPM	Achieved RPM 95% CI	Input TPM 95% CI	Output TPM 95% CI
32K input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-direct-aimd-20260 823-final	Confirmed bracket	12	12 to 12	492 to 516	48 to 48
Short input / long output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-direct-aimd-20260 823-final	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
32K input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Heterogeneous mixed	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / long output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable
Short input / short output	do-sixhour-aimd-202 60824-r1	Exploratory healthy observation	Not reported	Exploratory / CI unavailable	Exploratory / CI unavailable	Exploratory / CI unavailable

Exact confirmed-cell contract

Workload / run	Epochs / rows	Realized prompt tokens	Realized output tokens	Stream mode	Concurrency ceiling	Epoch duration
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 3	41-43	4	streamed	Not observed	5 s

Latency and service outcome

Workload / run	Success / scheduled	429 / 5xx / timeout	TTFT p50 seconds (95% epoch CI)	Latency p95 seconds (95% epoch CI)
Heterogeneous mixed do-direct-aimd-20260823-final	3 / 3	0 / 0 / 0	0.47 (0.42 to 0.53)	0.55 (0.49 to 0.62)

Independent two-minute soak

Each displayed offered rate applies only to the exact endpoint, workload recipe, and observed two-minute arrival interval. Headline RPM/TPM divide each arrival cohort by its full time through the last completion, including drain. Four contiguous analysis blocks use analysis_block_id as the exploratory interval unit; serial correlation is not modelled. This is not a longer-duration or time-of-day SLA.

Workload	Offered arrival rate	State / acceptance	Drain-inclusive cohort RPM mean (95% block t interval)	Drain-inclusive input TPM	Drain-inclusive output TPM
Heterogeneous mixed	Not soak-verified	Complete; measured fail	6 (6 to 6)	Not reported	Not reported
Short input / long output	Not soak-verified	Complete; measured fail	23.11 (20.46 to 25.77)	Not reported	Not reported
32K input / short output	Not soak-verified	Complete; measured fail	11.79 (11.12 to 12.46)	301,724	328.59
Short input / short output	1 RPS	Complete; pass	59.1 (56.23 to 61.97)	2,573	236.39

Post-overload recovery

Workload	Target / realized RPS	Recovery decision	Success rate (95% request CI)	TTFT / latency p95 seconds
Heterogeneous mixed	0.05 / 0.03	Pass	100.0% (20.7% to 100.0%)	0.44 / 0.49
Short input / long output	0.2 / 0.2	Measured fail	100.0% (61.0% to 100.0%)	0.84 / 4.89
32K input / short output	0.1 / 0.1	Measured fail	100.0% (43.9% to 100.0%)	2.5 / 2.93
Short input / short output	0.5 / 0.5	Pass	100.0% (79.6% to 100.0%)	1.22 / 1.39

Observed versus documented limits

Limit or capability	Documented	Max accepted / supported	Max retrieval-valid / realized	First explicit rejection	Censoring / finding
prompt context window	1000000	1004307	1004307	Not observed	Right censored
combined prompt + requested output	1000000	1002293	1002293	Not observed	Right censored
output limit	262144	262145	10380	Not observed	Realized generation observed
vision	True	observed_functional	observed_functional	Not observed	Functional in observed tests
tool calling	True	observed_functional	observed_functional	Not observed	Functional in observed tests
structured output	True	observed_functional	observed_functional	Not observed	Functional in observed tests

Low-load and capability-task quality

Exact task / workload	Run / phase / offered load	Scored requests	Pass rate	95% request CI
Short input / short output direct-short-1068607726095	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-111787537276	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-119868865272	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-146993401031	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-186039080954	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-230010608576	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-266132357919	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-284404008718	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-287255035368	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-301834412153	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-309131239084	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%
Short input / short output direct-short-329015823457	do-direct-aimd-20260823-final Additive aimd	1	100.0%	20.7% to 100.0%

Every displayed percentage is one exact source, task, phase, offered rate, stream mode, and token-target stratum; AIMD phases and rates are never pooled. Service failure and task failure are separate: an unavailable or throttled response reduces service goodput, while an HTTP-successful but incorrect answer reduces quality-adjusted goodput.

Paired quality change under load

Exact-payload low-load and near-load pairs are averaged within their predeclared analysis block. The displayed Student-t interval then uses the independent analysis_block_id means as its units; both block and pair counts are shown.

Workload / run	Blocks / pairs	Mean near-minus-low score	95% t interval across block means	Pair acceptance
32K input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Heterogeneous mixed do-direct-soak-20260823-r1	4 / 5	0.25	-0.55 to 1.05	3 / 5
Short input / long output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	0 / 4
Short input / short output do-direct-soak-20260823-r1	4 / 4	0	0 to 0	4 / 4

6. Study-wide limitations

- This is an observed operating envelope for the tested account, client region, time period, request mix, and API version; it is not a universal SLA.
- Prompt tokens divided by TTFT is an end-to-end prefill proxy because the service does not expose direct server-side prefill timing.
- A request limit being accepted is different from the model actually realizing that many output tokens.
- Capacity is confirmed only when at least three valid healthy epochs exist at the same offered rate. Otherwise the highest healthy rate is exploratory.
- p99 is suppressed below 1,000 relevant observations; sparse p95 values are labelled exploratory.
- Quality comparisons are endpoint observations, not causal DigitalOcean failures, unless a matched external-provider control is supplied separately.
- Streaming events are not token timestamps. The legacy first-to-last SSE-event rate is audit-only; valid endpoint comparisons use aggregate wall-clock goodput or the explicitly labelled post-TTFT end-to-end proxy.
- Buffered responses do not expose token-level TTFT. Their full-response arrival times are retained as latency but censored from TTFT and prefill summaries.

Public artifact map

Artifact	Present	Purpose
normalized-requests.csv	Yes	Sanitized request-level metrics; no prompt or response text.
normalized-epochs.csv	Yes	Independent load-epoch results used for capacity intervals.
endpoint-summary.csv	Yes	One-row engineering summary per exact endpoint.
endpoint-workload-metrics.csv	Yes	Workload-specific latency, throughput, reliability, and quality.
capacity-summary.csv	Yes	AIMD confirmation points, brackets, and right-censoring; no automatic headroom.
soak-cell-summary.csv	Yes	Exact two-minute soak result per endpoint and workload; separate from AIMD.
soak-block-summary.csv	Yes	Four predeclared 30-second analysis_block_id units per complete soak cell.
quality-pair-summary.csv	Yes	Exact-payload low-load versus near-load quality-pair evidence.
recovery-summary.csv	Yes	Post-soak recovery phases and predeclared pass/failure reasons.
coverage-ledger.csv	Yes	Every planned cell and its terminal evidence state.
coverage-matrix.csv	Yes	Compact endpoint-by-dimension coverage view.
scope-exclusions.csv	Yes	Manifest-declared subtests that remain explicitly untested.
observed-limits.csv	Yes	Observed versus documented boundary results.
metric-audit.csv	Yes	Request-traceable legacy versus corrected timing metrics; no trimming.
cache-state-metrics.csv	Yes	TTFT and prefill proxies split by observed hit, miss, or unknown cache state.
capability-evidence.csv	Yes	Transport support, functional correctness, and malformed-input validation kept separate.
charts/	Yes	Derived, publication-safe figures used in this PDF.

Public-data boundary

Plain language: This PDF contains performance evidence, not test content or account details.

Technical definition: The builder reads only analysis.json (or its gzip-compressed equivalent), 16 named derived CSVs, and declared chart images. Final builds require the matching manifest and a passing public-bundle safety scan.